

BFS Traversal

Step 1: Start at E

Legend

Complete

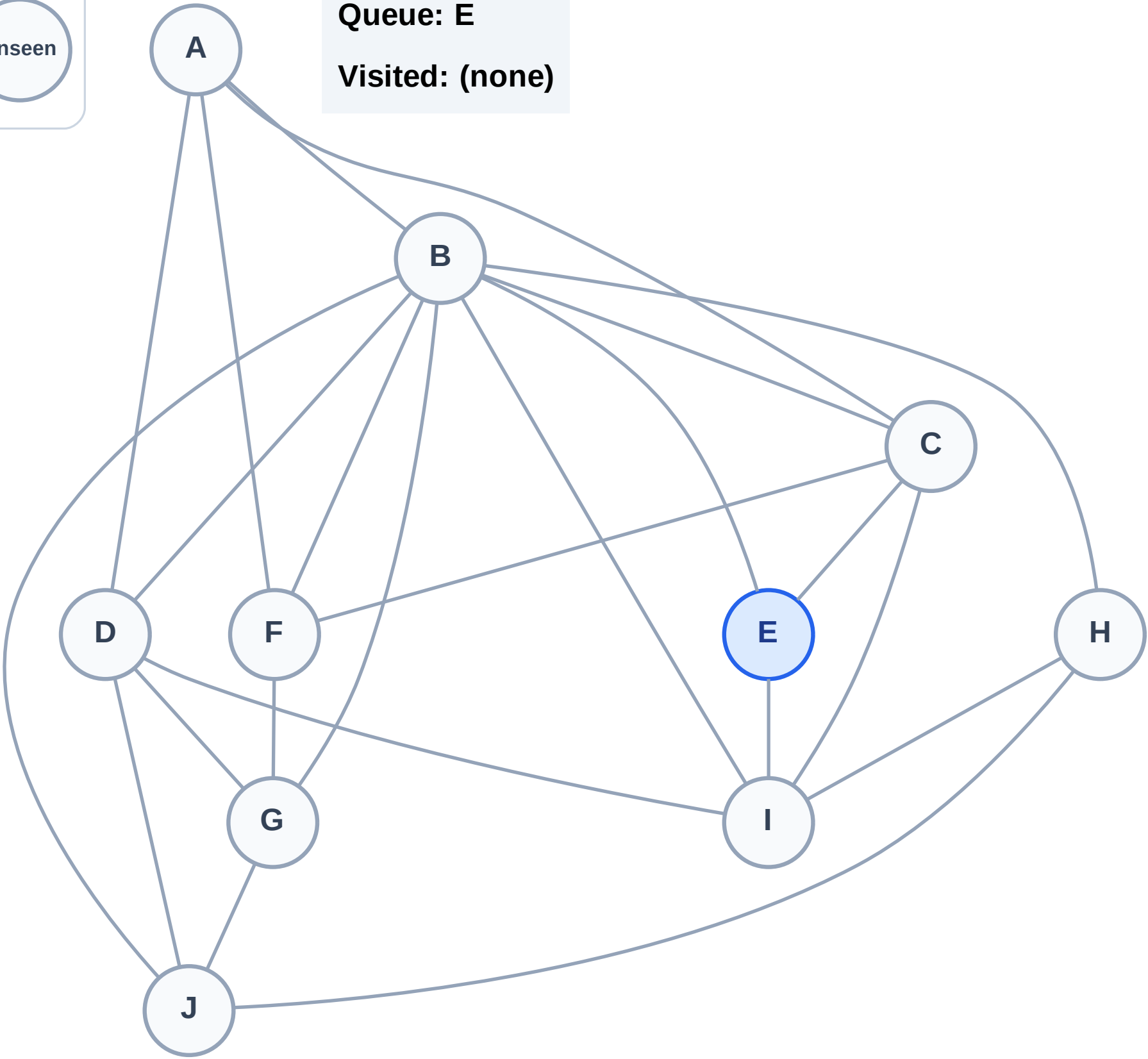
Processing

Discovered

Unseen

Queue: E

Visited: (none)



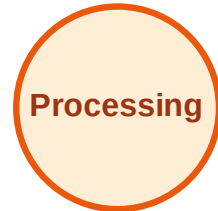
BFS Traversal

Step 2: Process node E

Legend



Complete



Processing



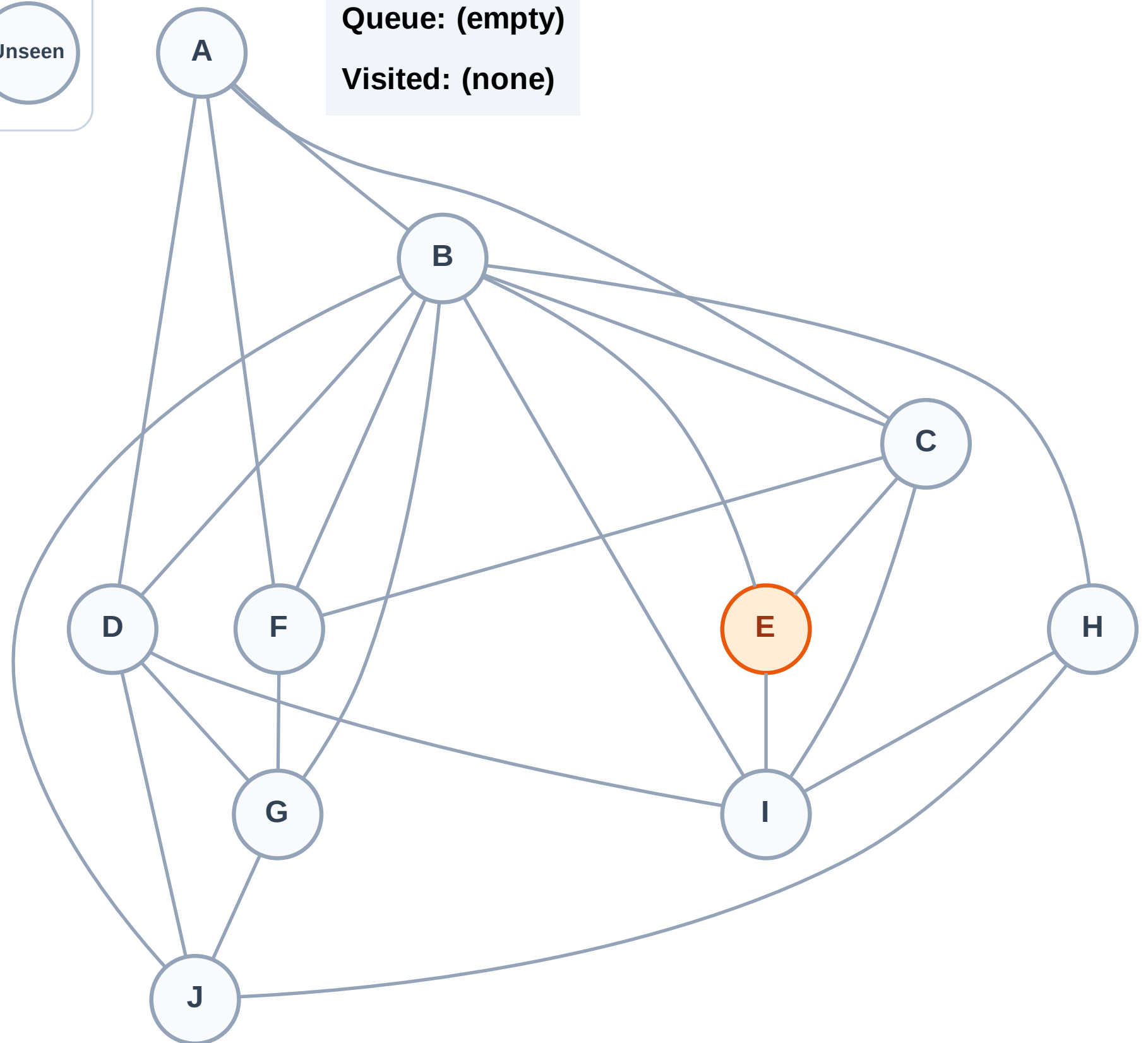
Discovered



Unseen

Queue: (empty)

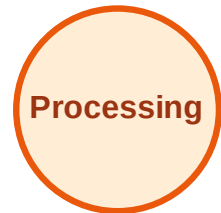
Visited: (none)



BFS Traversal

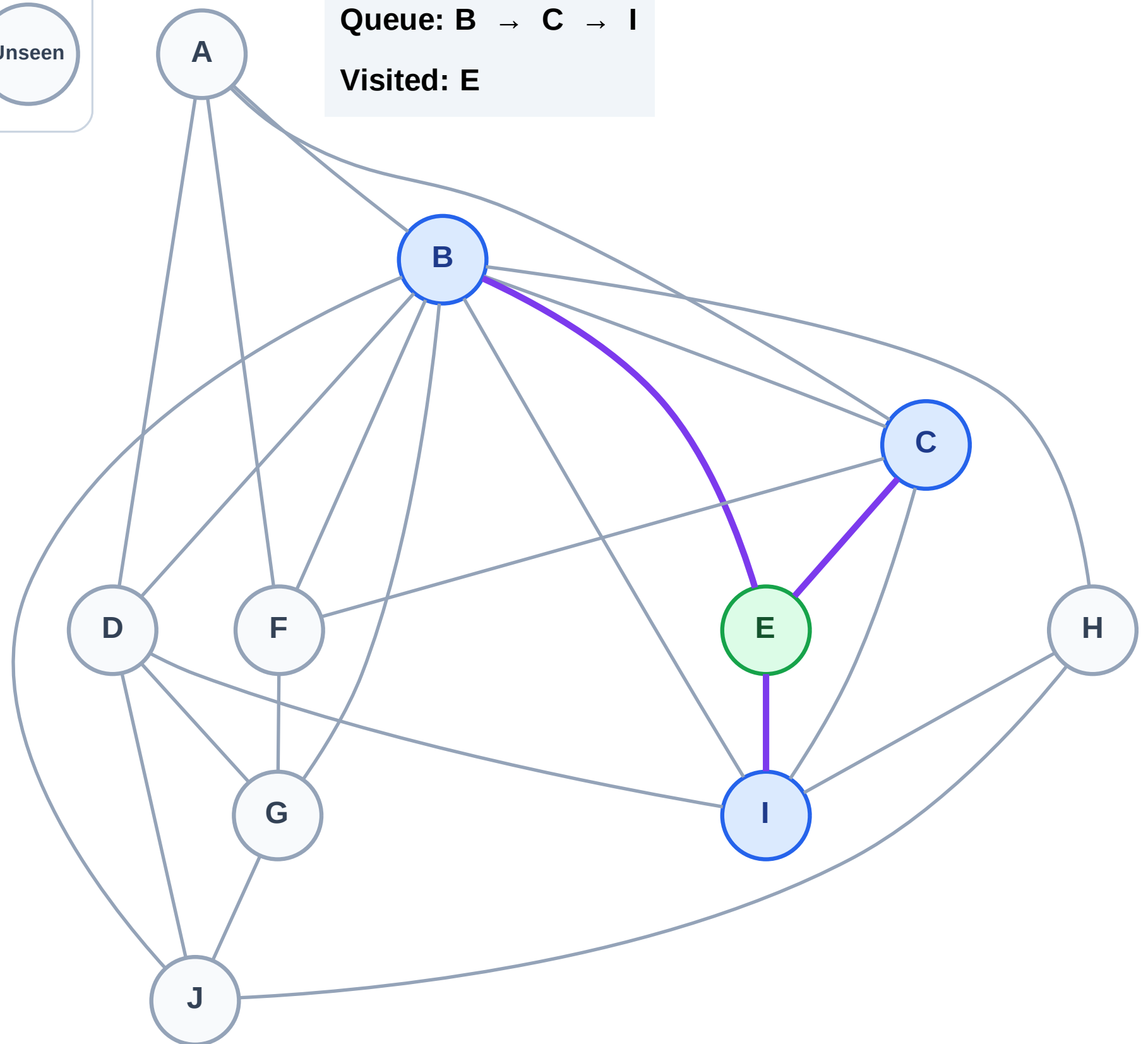
Step 3: Discover B, C, I from E

Legend



Queue: B → C → I

Visited: E



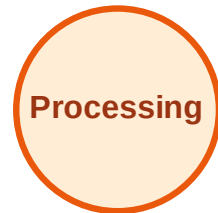
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



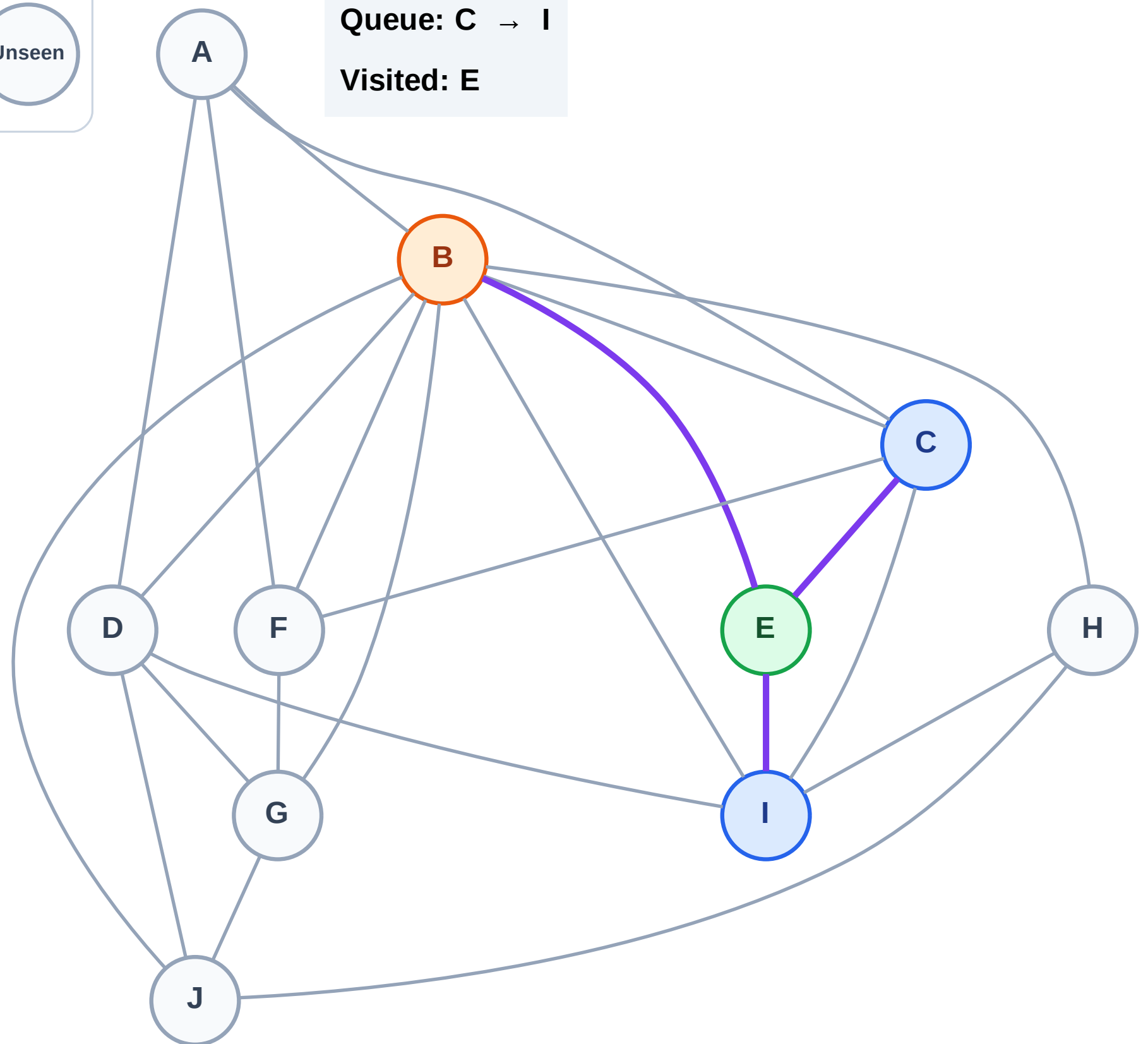
Discovered



Unseen

Queue: C → I

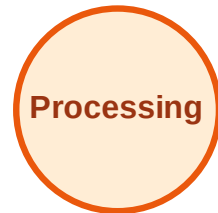
Visited: E



BFS Traversal

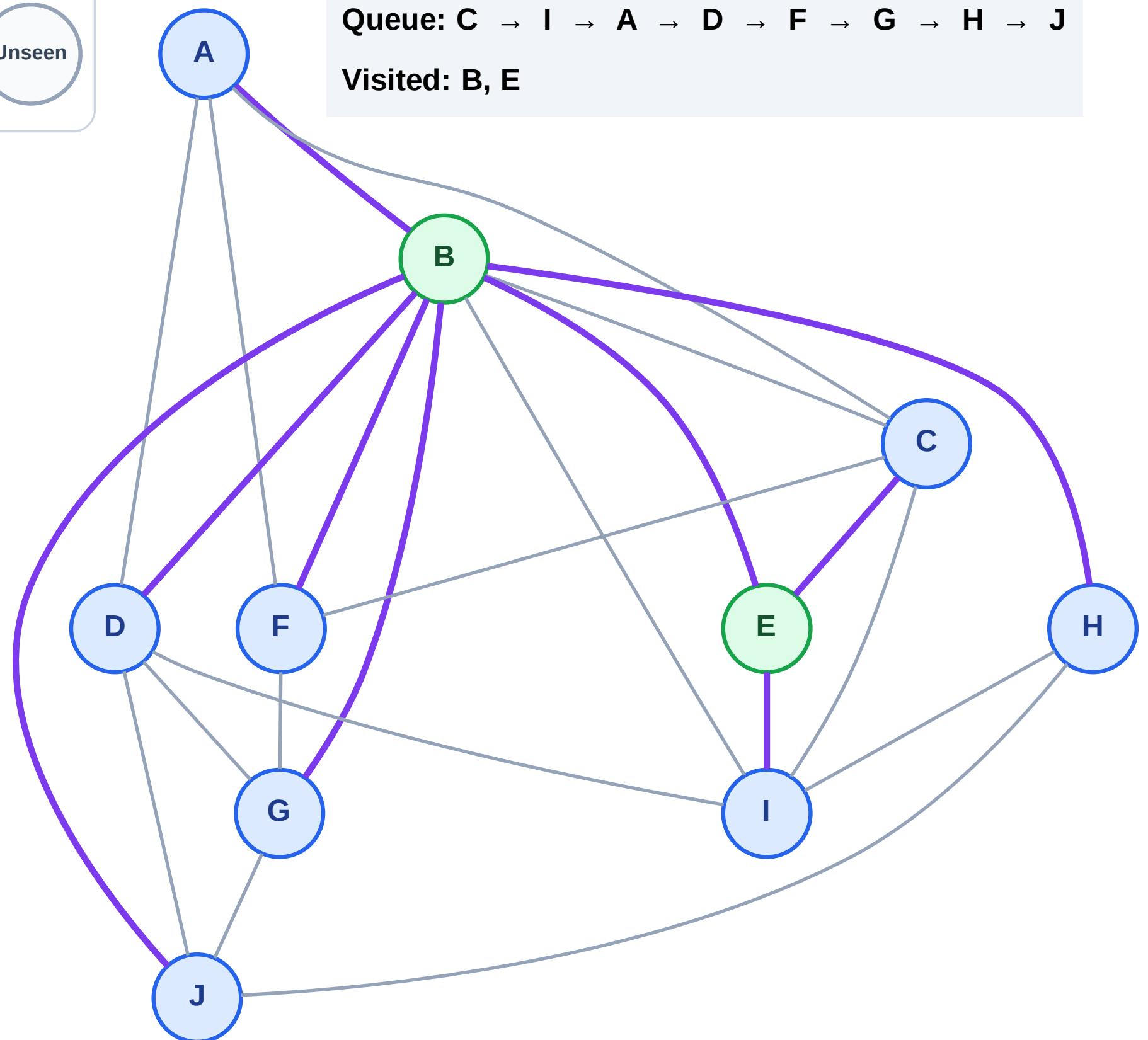
Step 5: Discover A, D, F, G, H, J from B

Legend



Queue: C → I → A → D → F → G → H → J

Visited: B, E



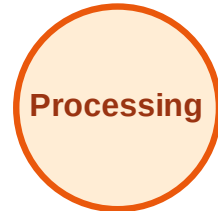
BFS Traversal

Step 6: Process node C

Legend



Complete



Processing



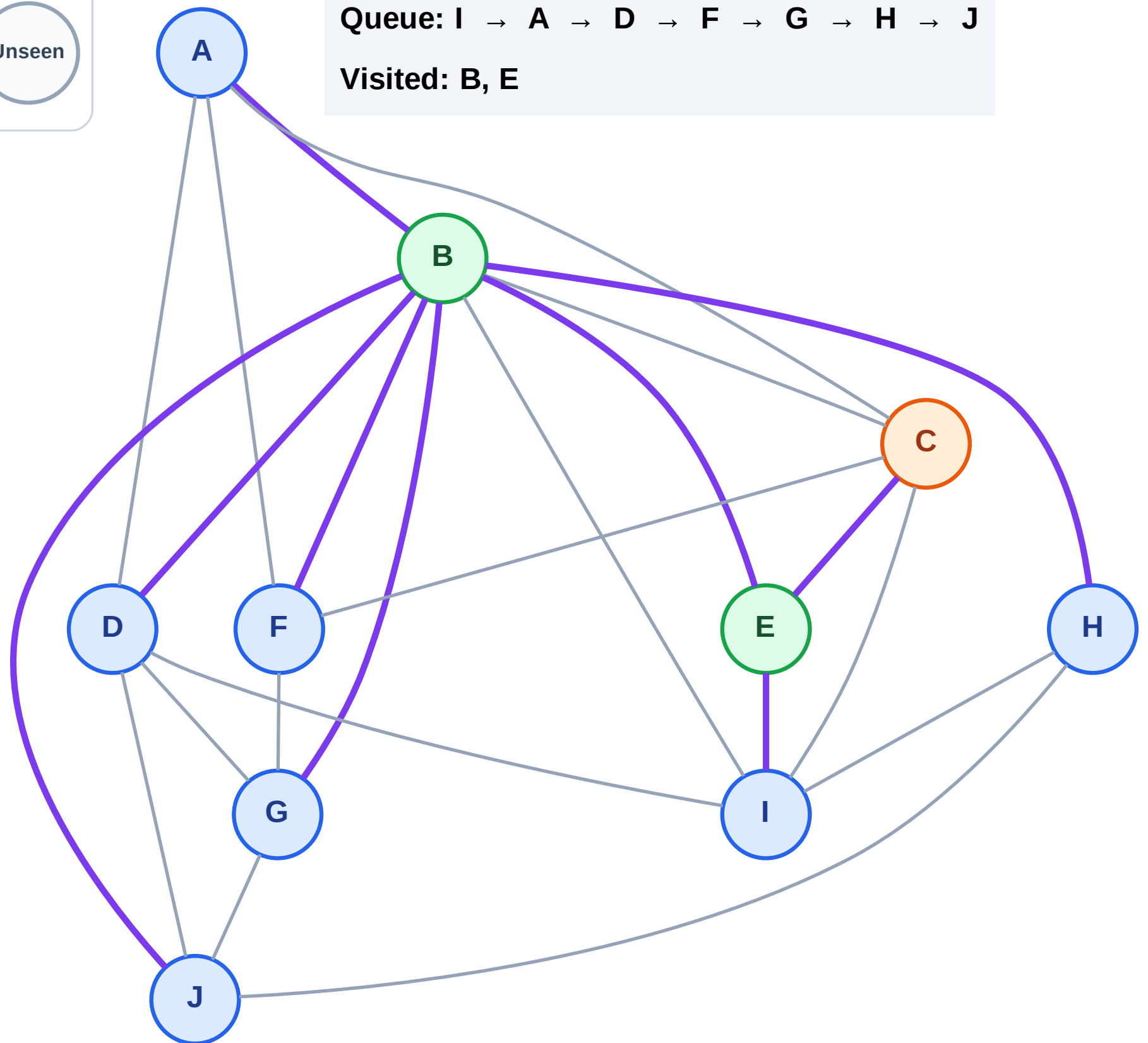
Discovered



Unseen

Queue: I → A → D → F → G → H → J

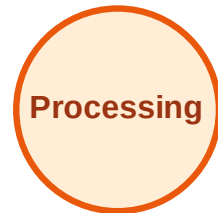
Visited: B, E



BFS Traversal

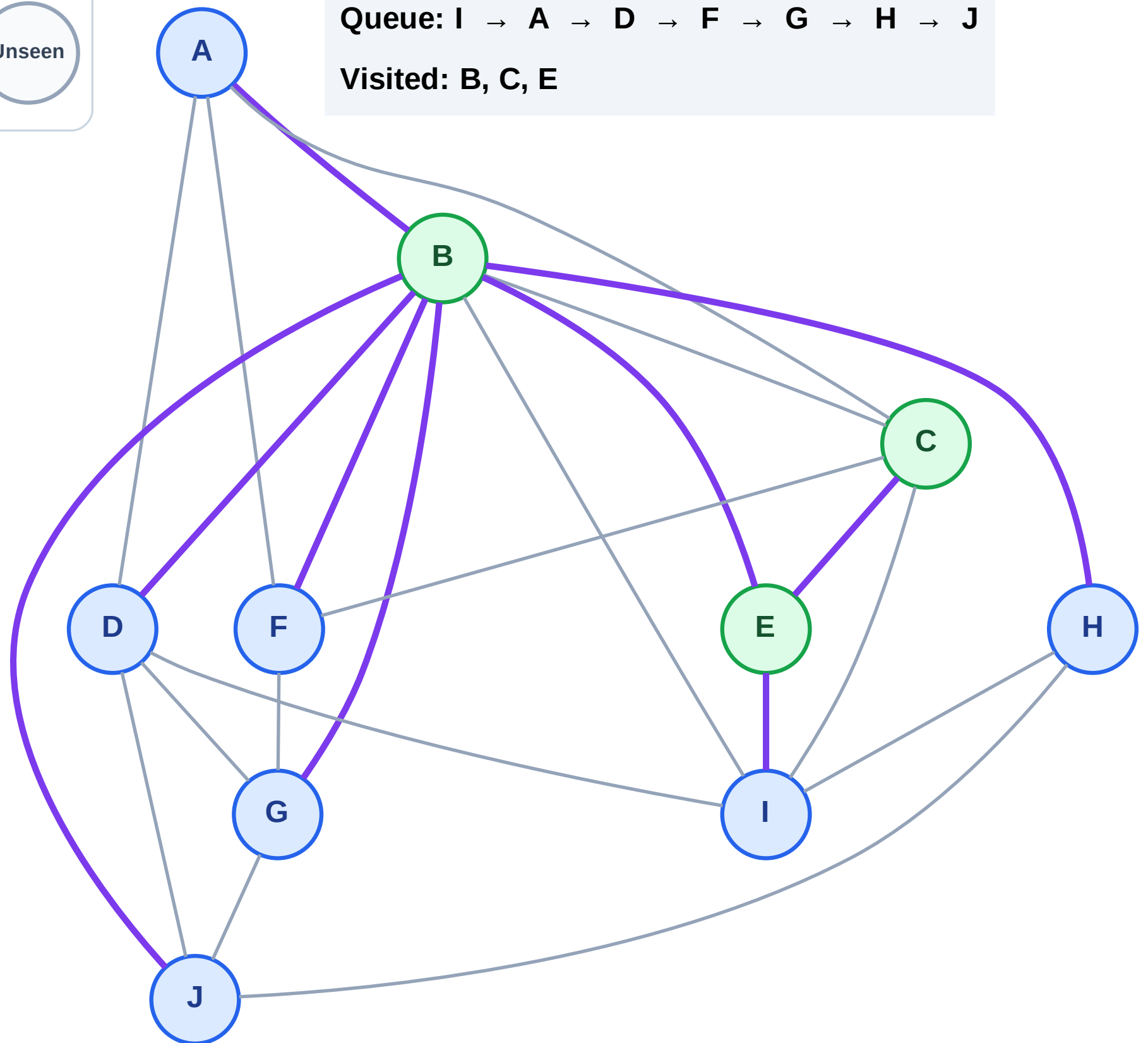
Step 7: No new neighbors from C

Legend



Queue: I → A → D → F → G → H → J

Visited: B, C, E



BFS Traversal

Step 8: Process node I

Legend

Complete

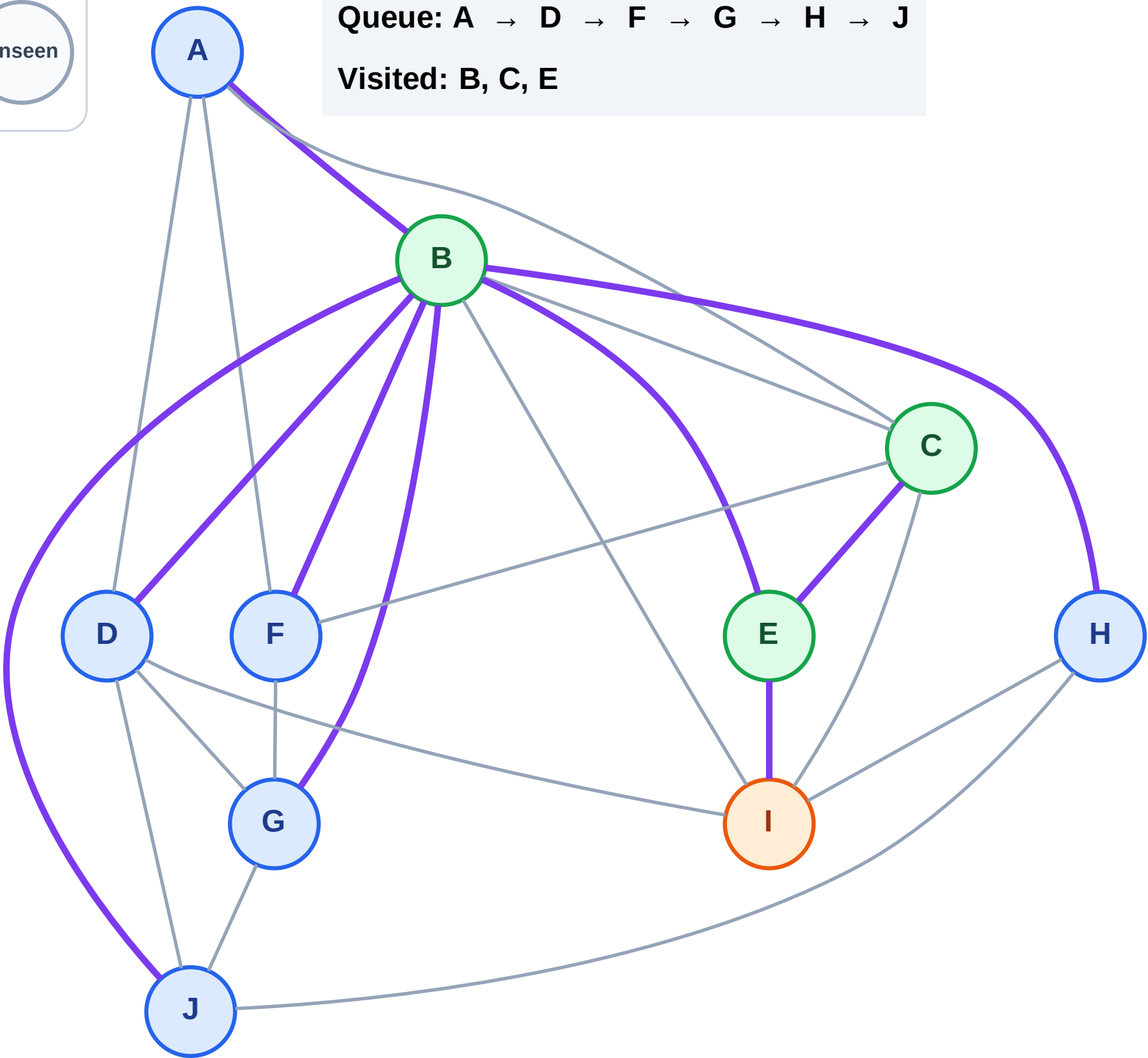
Processing

Discovered

Unseen

Queue: A → D → F → G → H → J

Visited: B, C, E



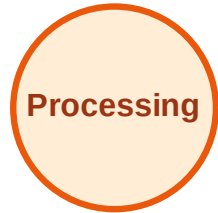
BFS Traversal

Step 9: No new neighbors from I

Legend



Complete



Processing



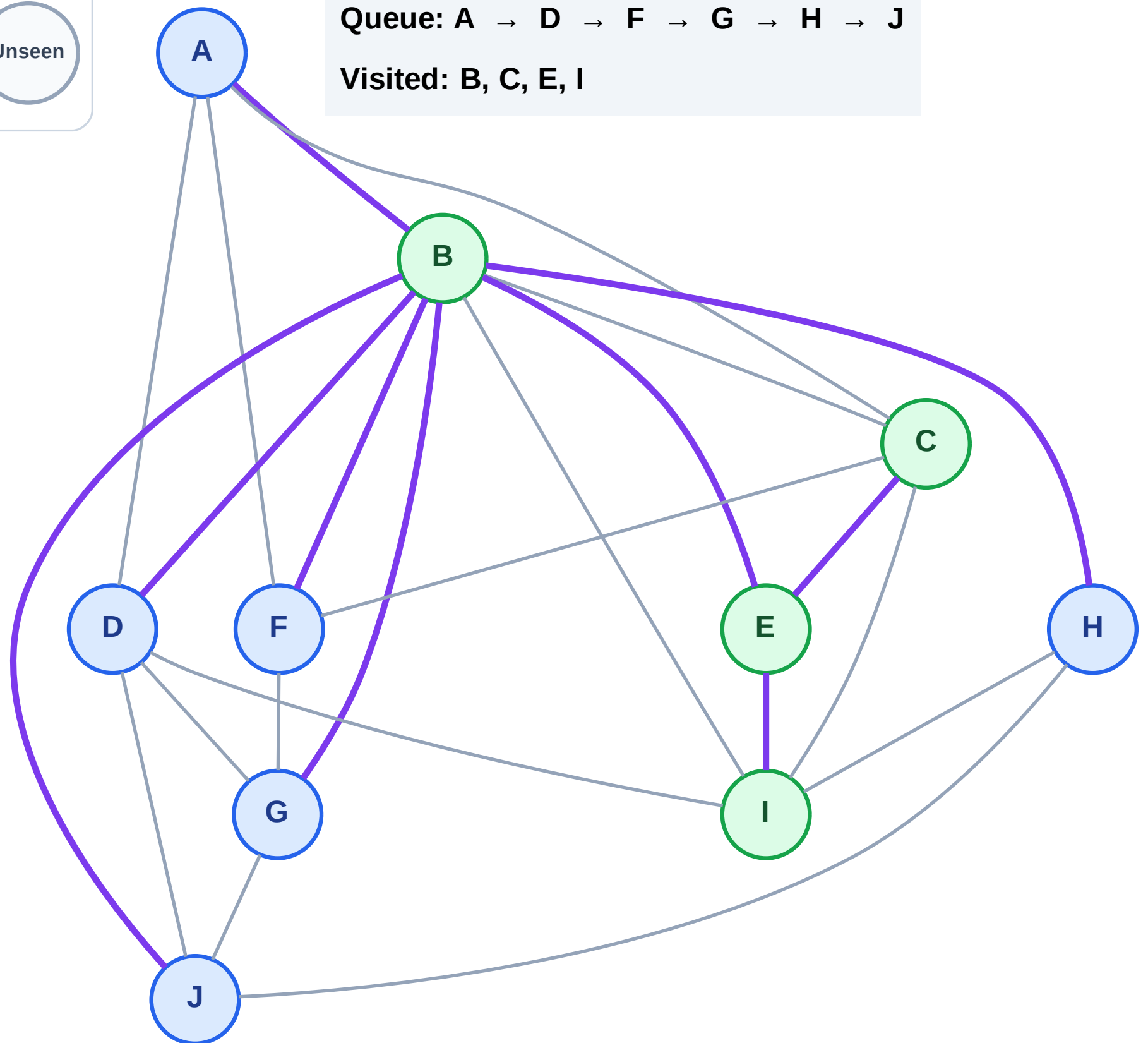
Discovered



Unseen

Queue: A → D → F → G → H → J

Visited: B, C, E, I



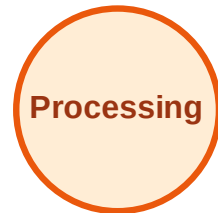
BFS Traversal

Step 10: Process node A

Legend



Complete



Processing



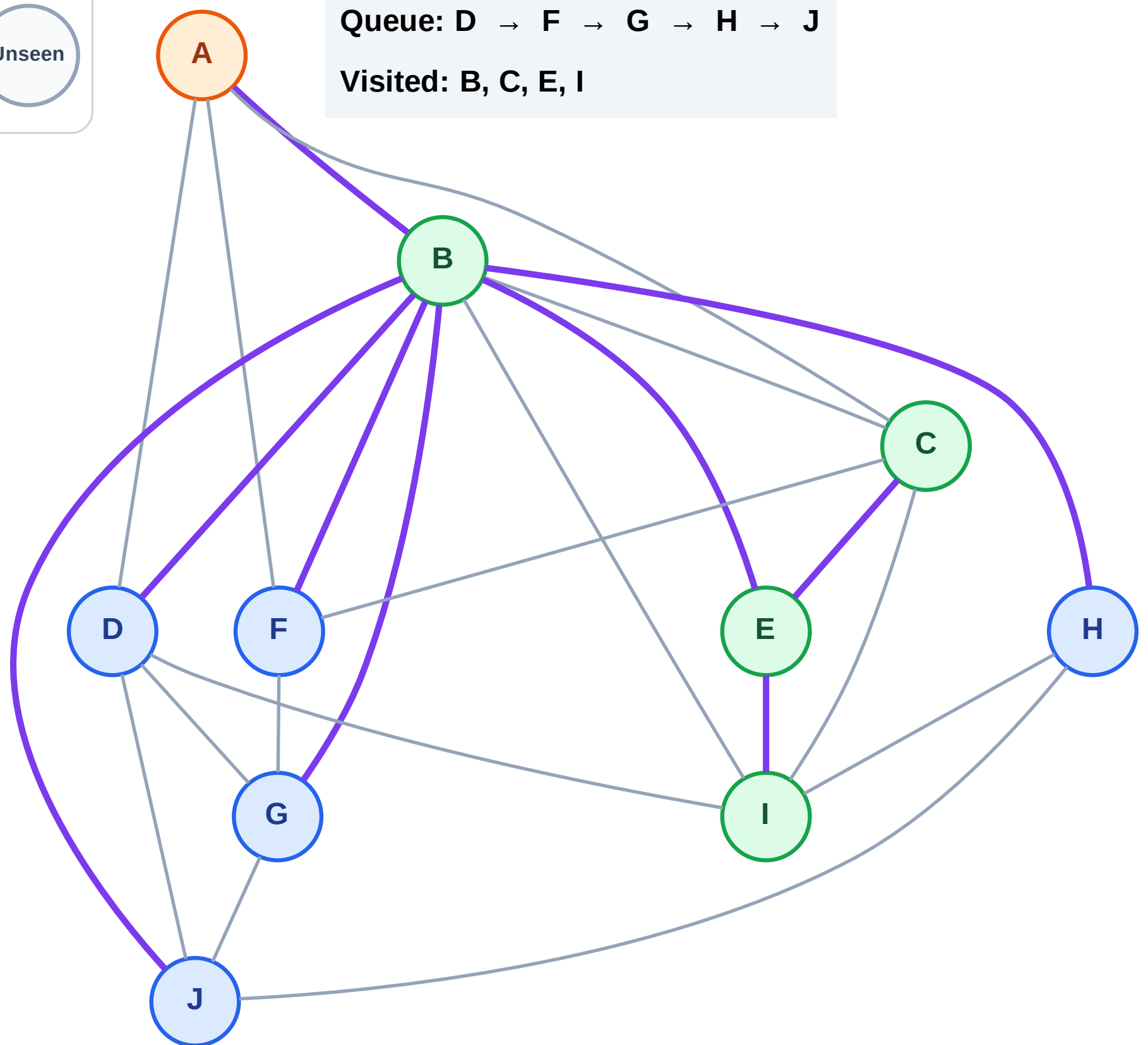
Discovered



Unseen

Queue: D → F → G → H → J

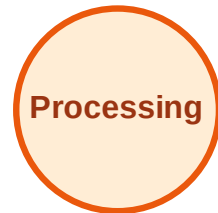
Visited: B, C, E, I



BFS Traversal

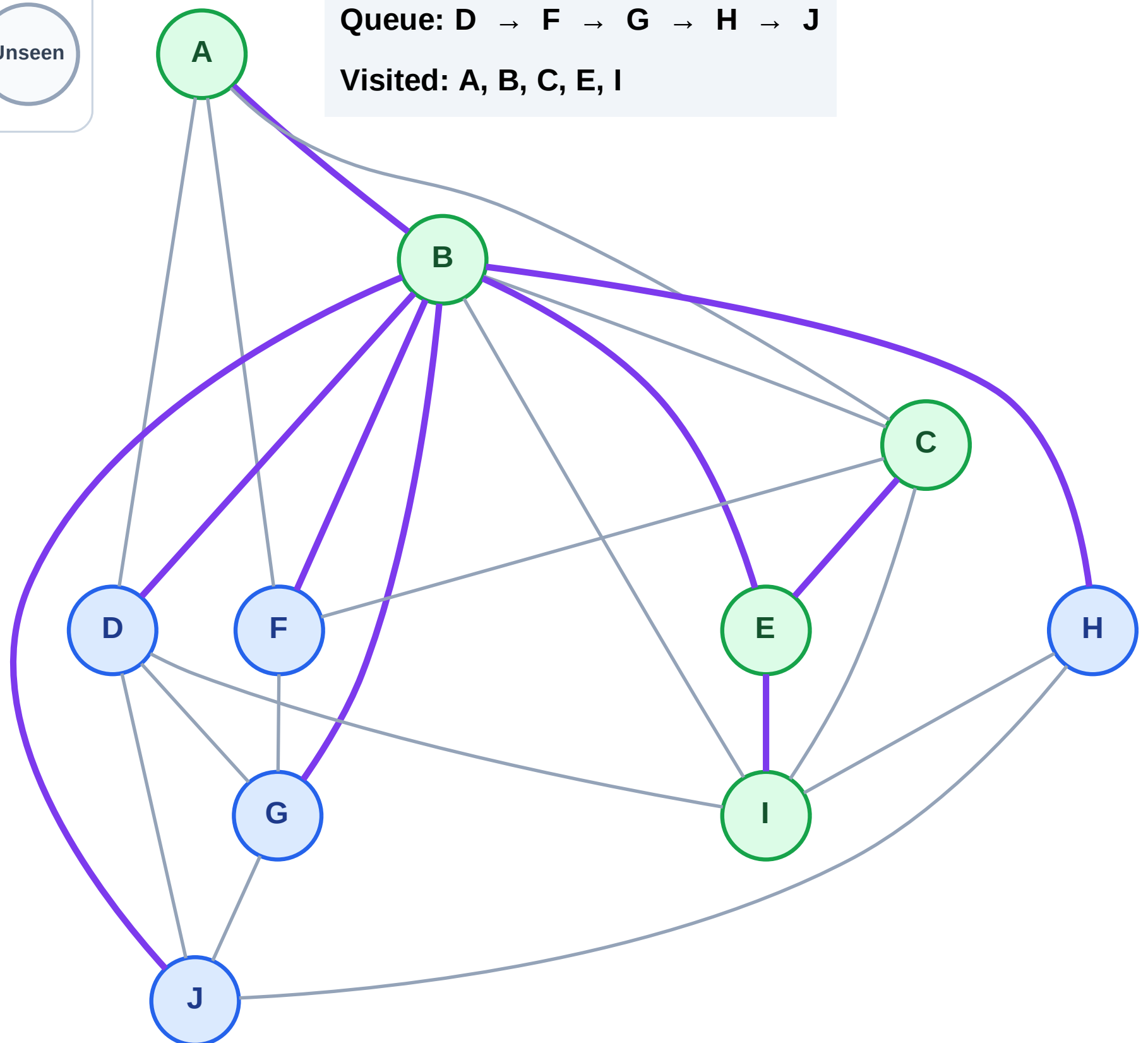
Step 11: No new neighbors from A

Legend



Queue: D → F → G → H → J

Visited: A, B, C, E, I



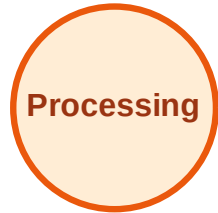
BFS Traversal

Step 12: Process node D

Legend



Complete



Processing



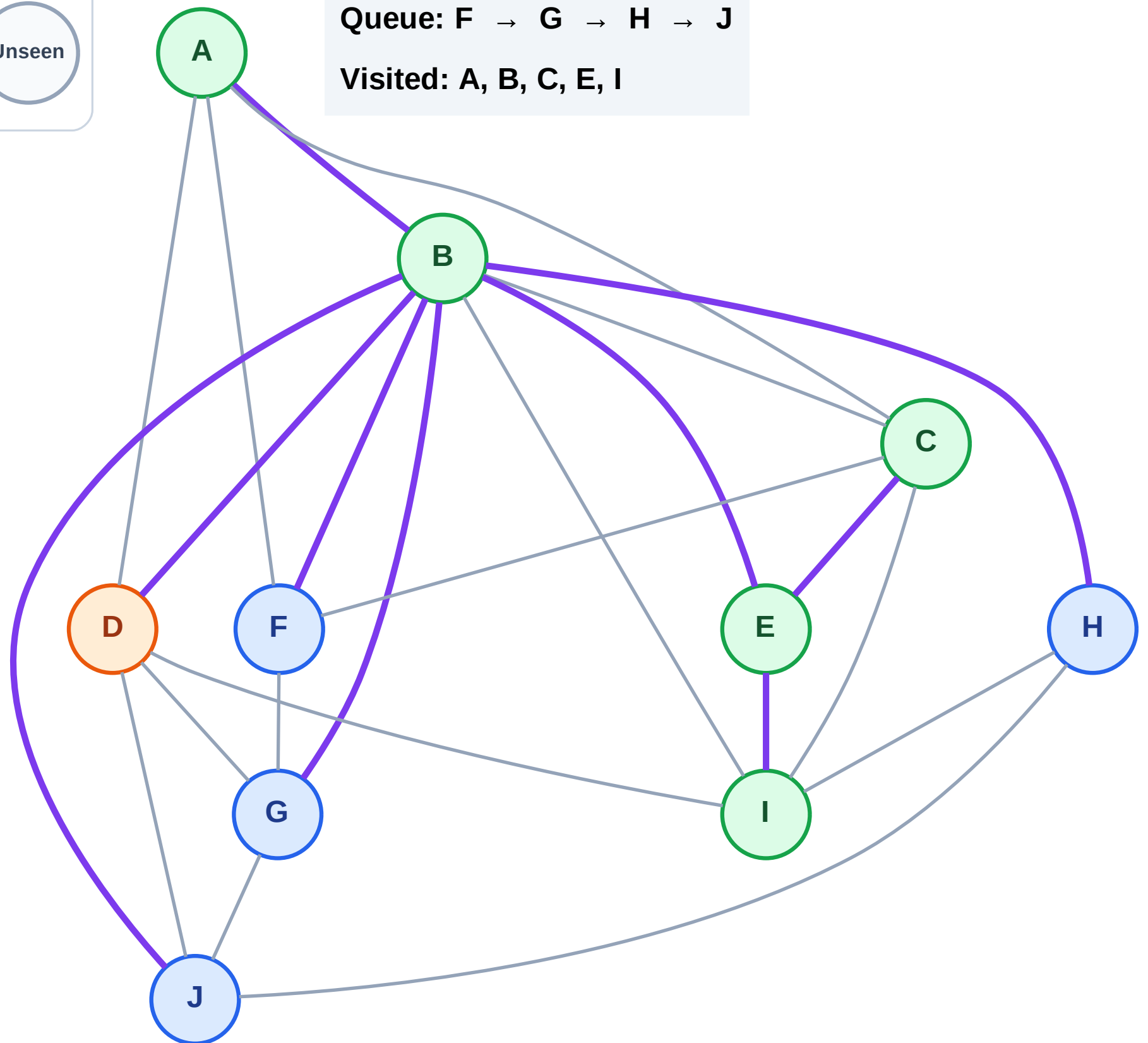
Discovered



Unseen

Queue: F → G → H → J

Visited: A, B, C, E, I



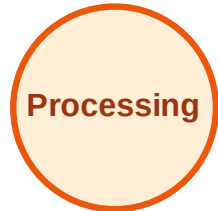
BFS Traversal

Step 13: No new neighbors from D

Legend



Complete



Processing



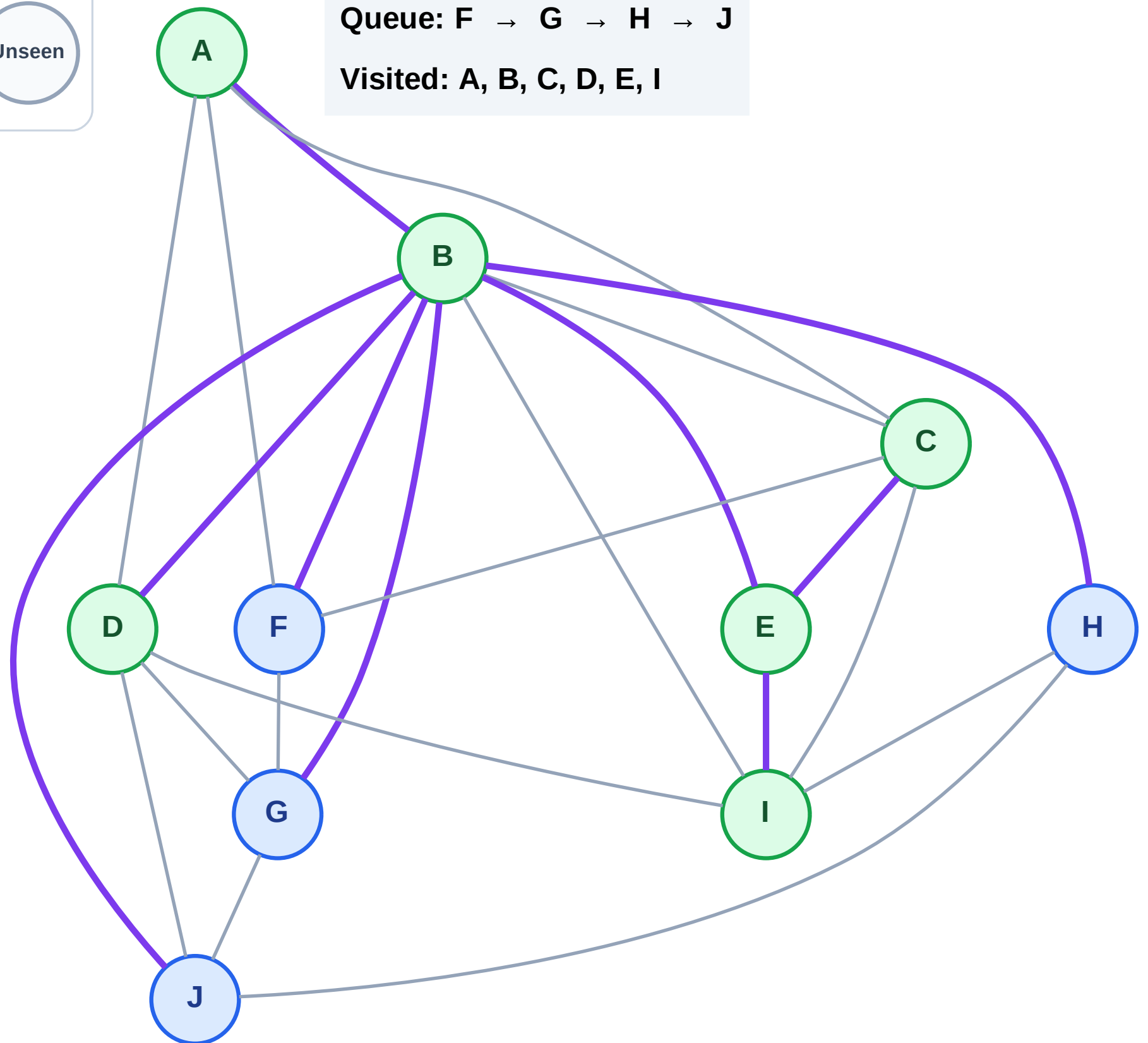
Discovered



Unseen

Queue: F → G → H → J

Visited: A, B, C, D, E, I



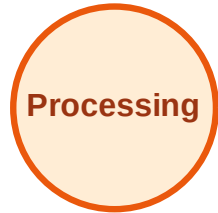
BFS Traversal

Step 14: Process node F

Legend



Complete



Processing



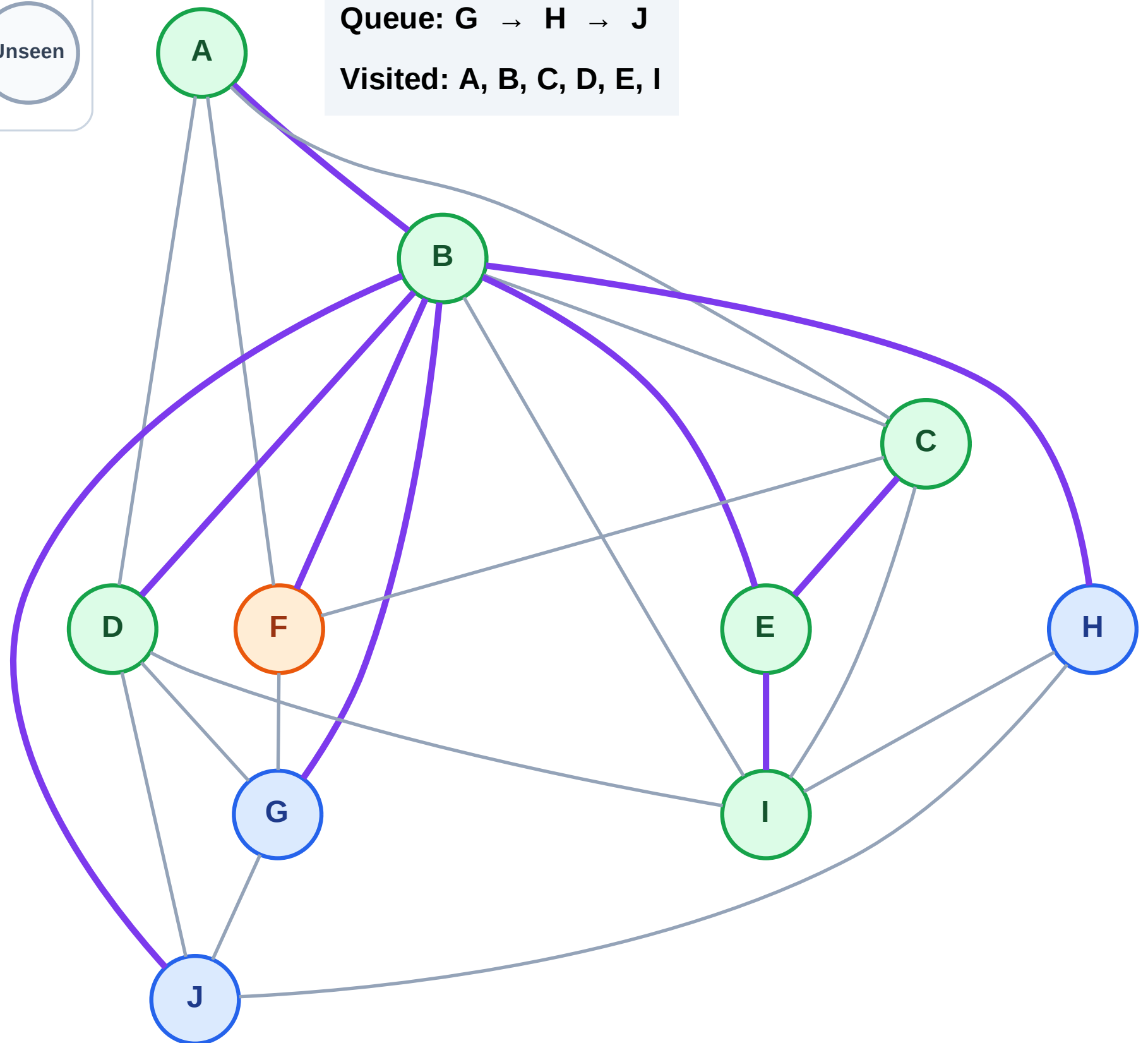
Discovered



Unseen

Queue: G → H → J

Visited: A, B, C, D, E, I



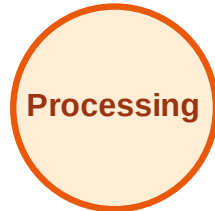
BFS Traversal

Step 15: No new neighbors from F

Legend



Complete



Processing



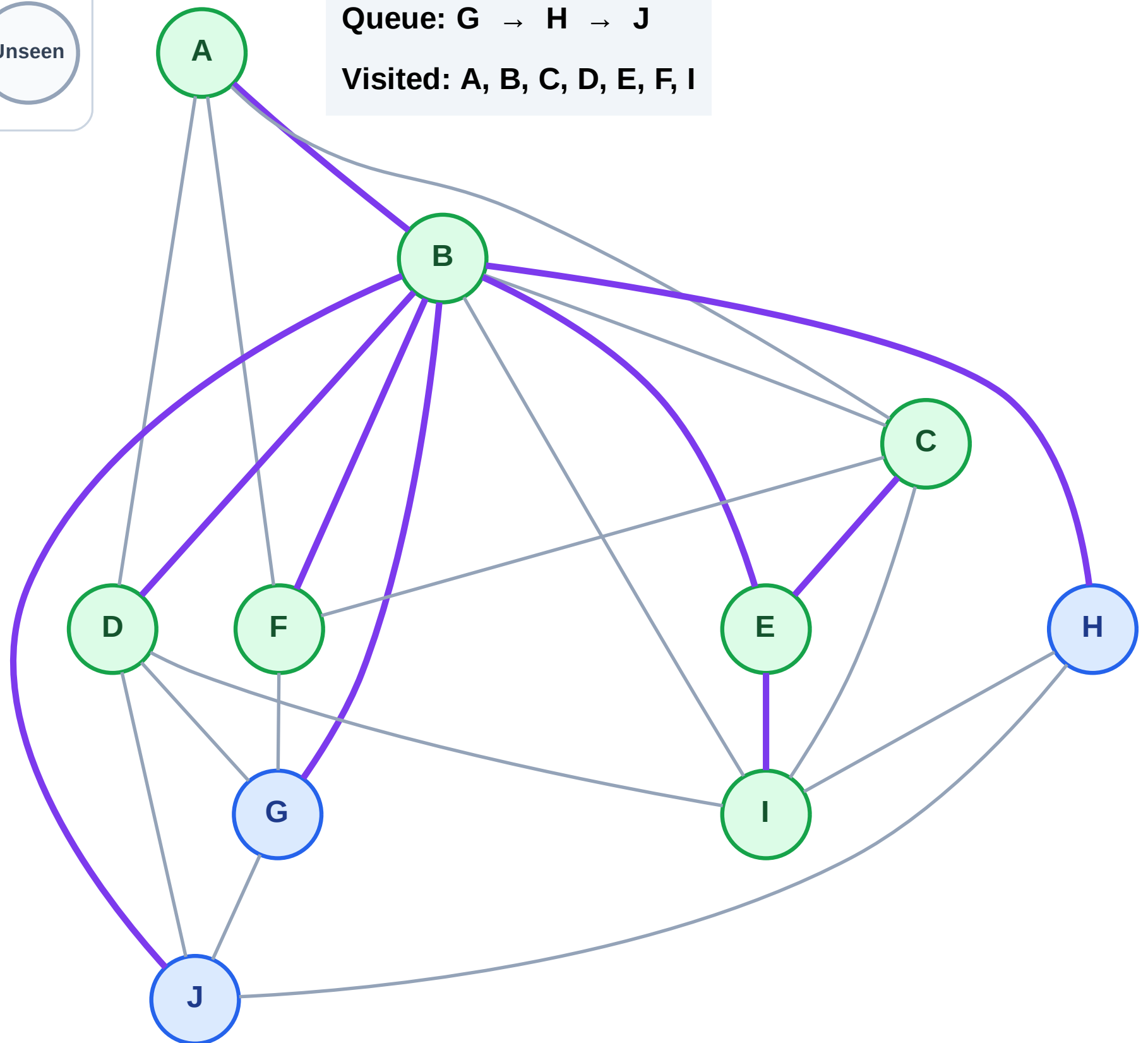
Discovered



Unseen

Queue: G → H → J

Visited: A, B, C, D, E, F, I



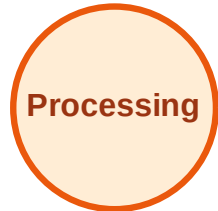
BFS Traversal

Step 16: Process node G

Legend



Complete



Processing



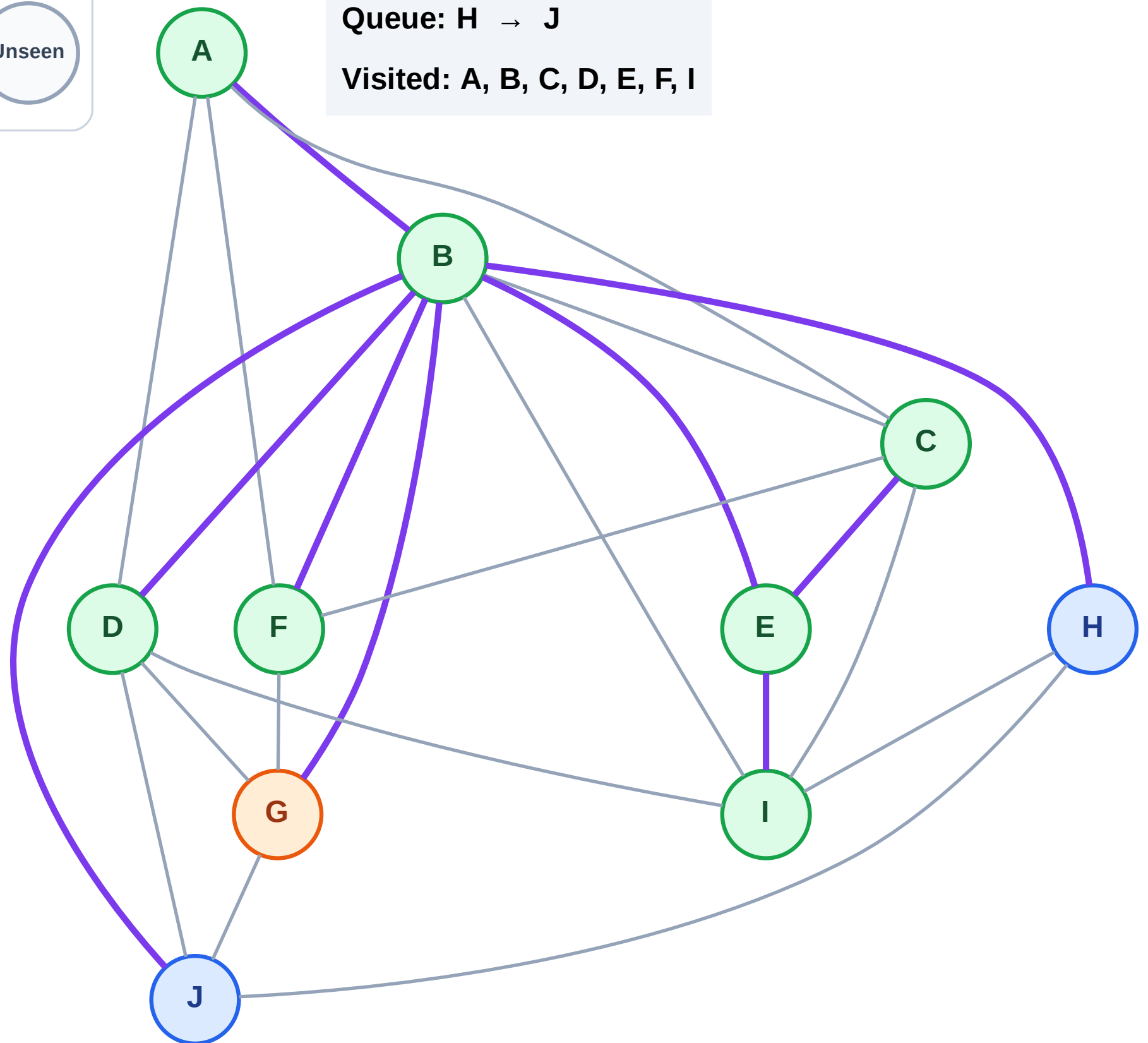
Discovered



Unseen

Queue: H → J

Visited: A, B, C, D, E, F, I



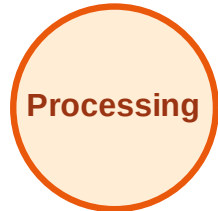
BFS Traversal

Step 17: No new neighbors from G

Legend



Complete



Processing



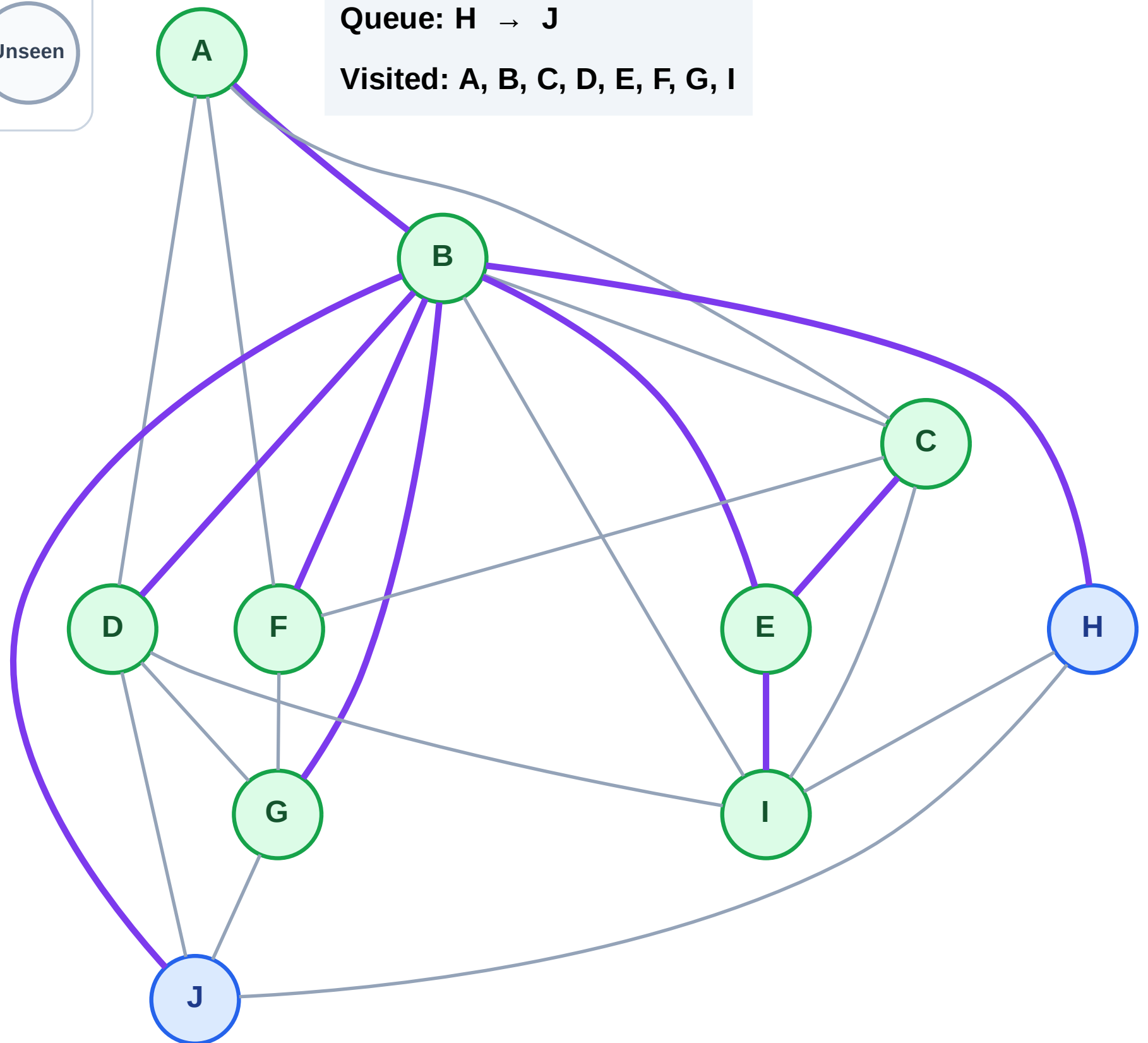
Discovered



Unseen

Queue: H → J

Visited: A, B, C, D, E, F, G, I



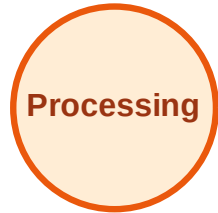
BFS Traversal

Step 18: Process node H

Legend



Complete



Processing



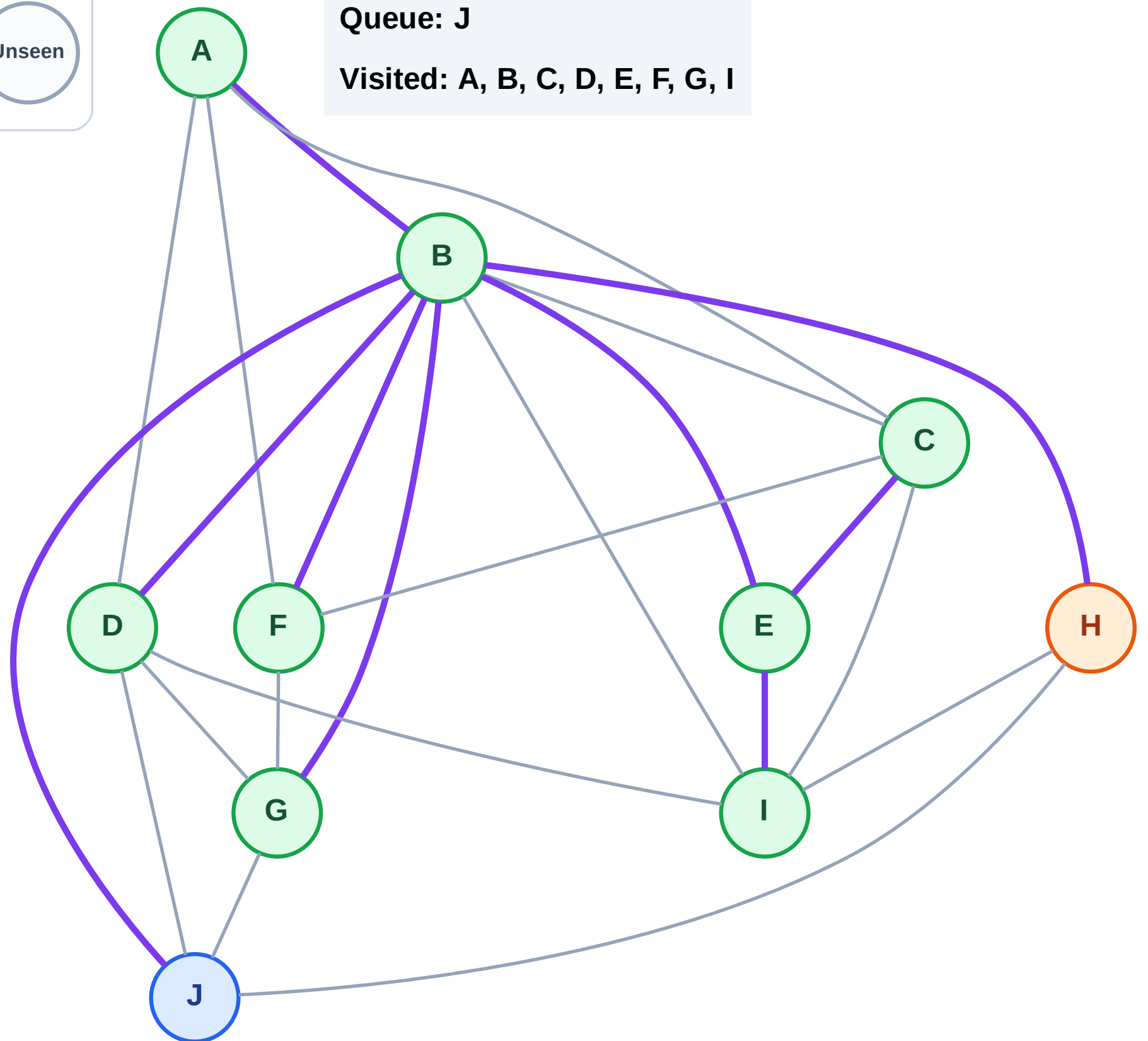
Discovered



Unseen

Queue: J

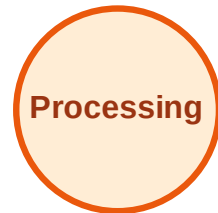
Visited: A, B, C, D, E, F, G, I



BFS Traversal

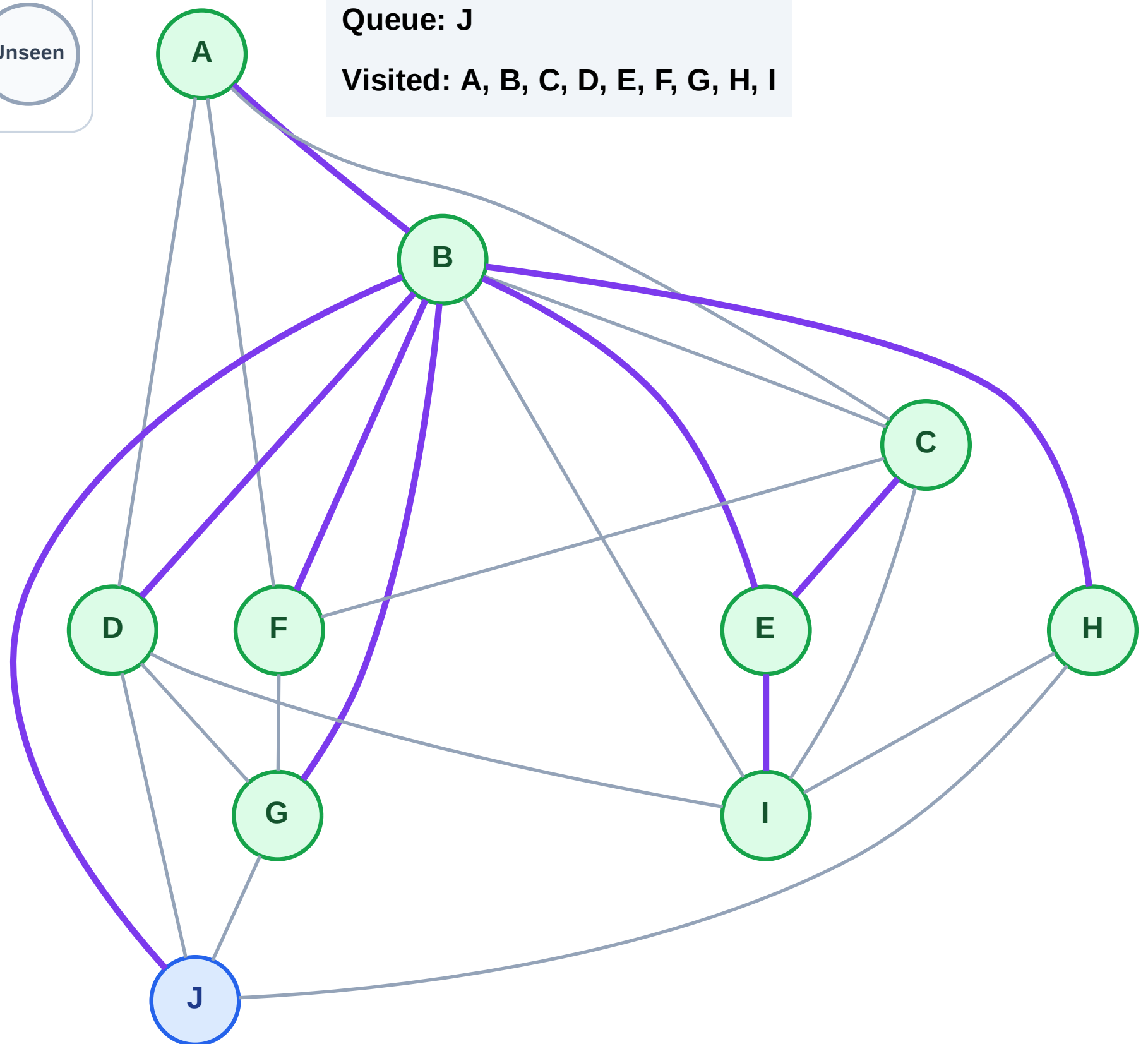
Step 19: No new neighbors from H

Legend



Queue: J

Visited: A, B, C, D, E, F, G, H, I



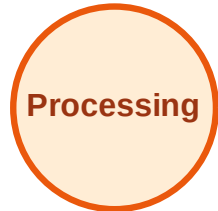
BFS Traversal

Step 20: Process node J

Legend



Complete



Processing



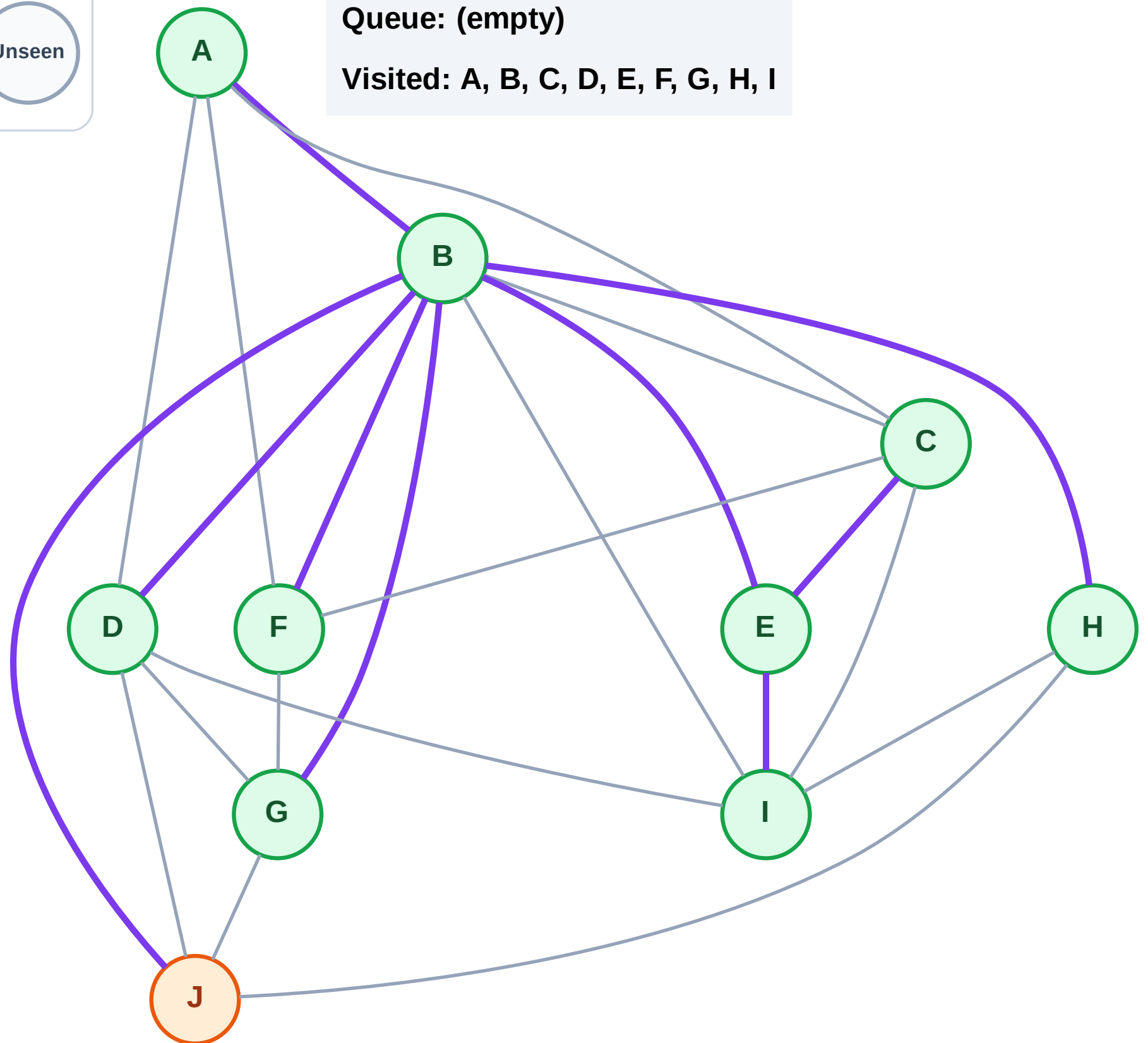
Discovered



Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I



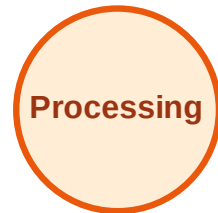
BFS Traversal

Step 21: No new neighbors from J

Legend



Complete



Processing



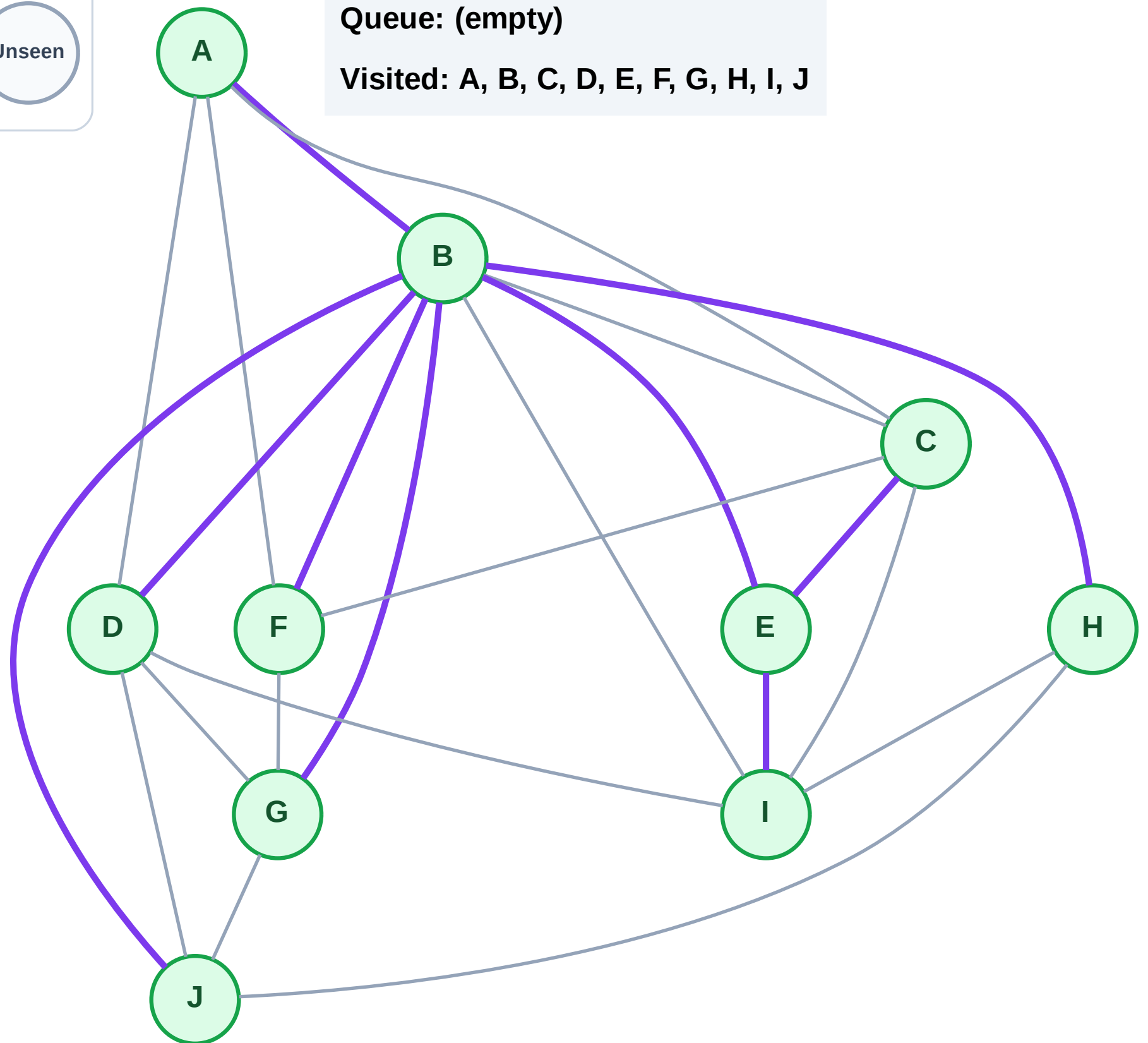
Discovered



Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



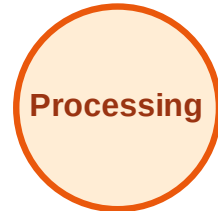
DFS Traversal

Step 1: Start at E

Legend



Complete



Processing



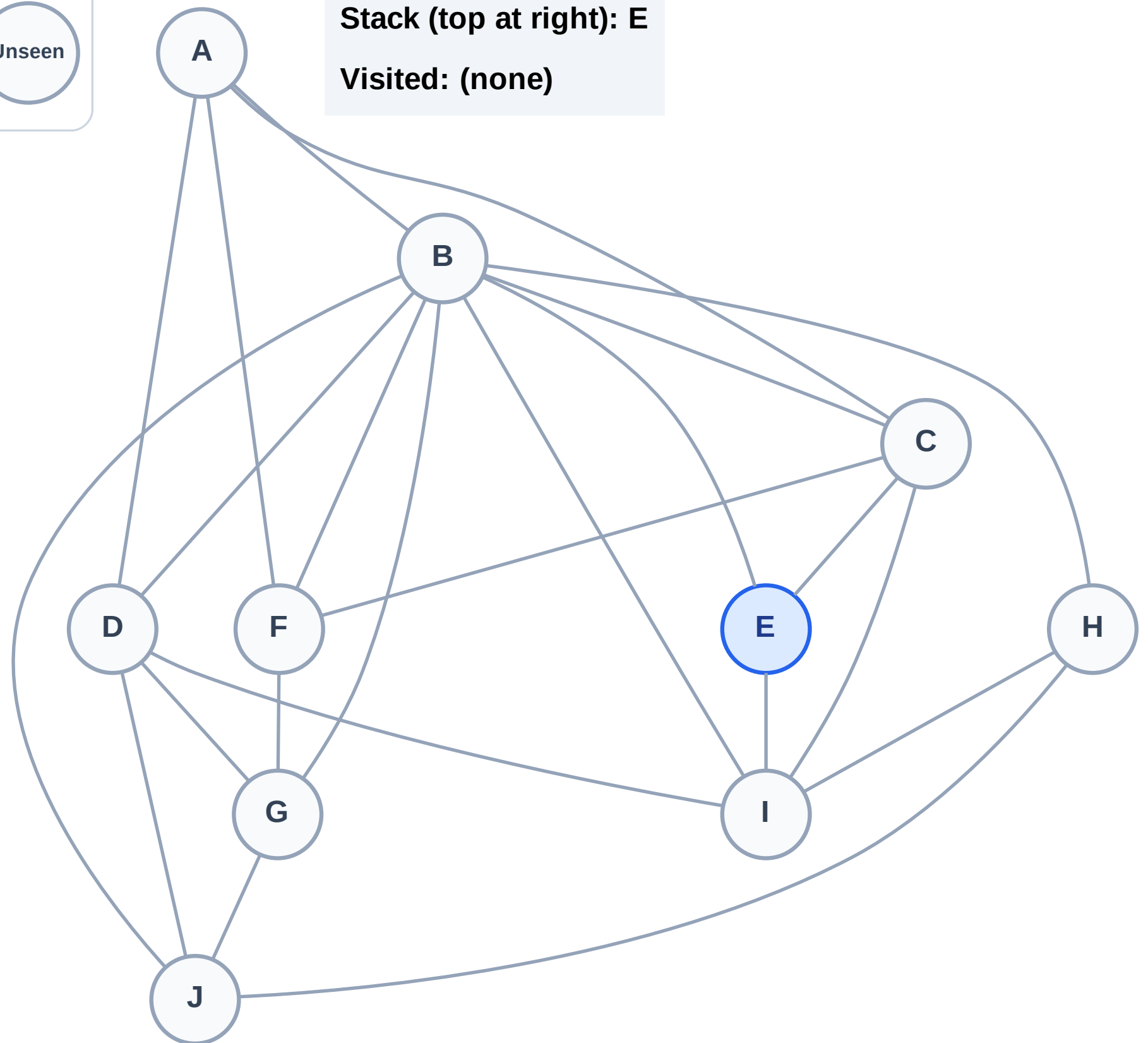
Discovered



Unseen

Stack (top at right): E

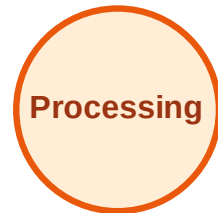
Visited: (none)



DFS Traversal

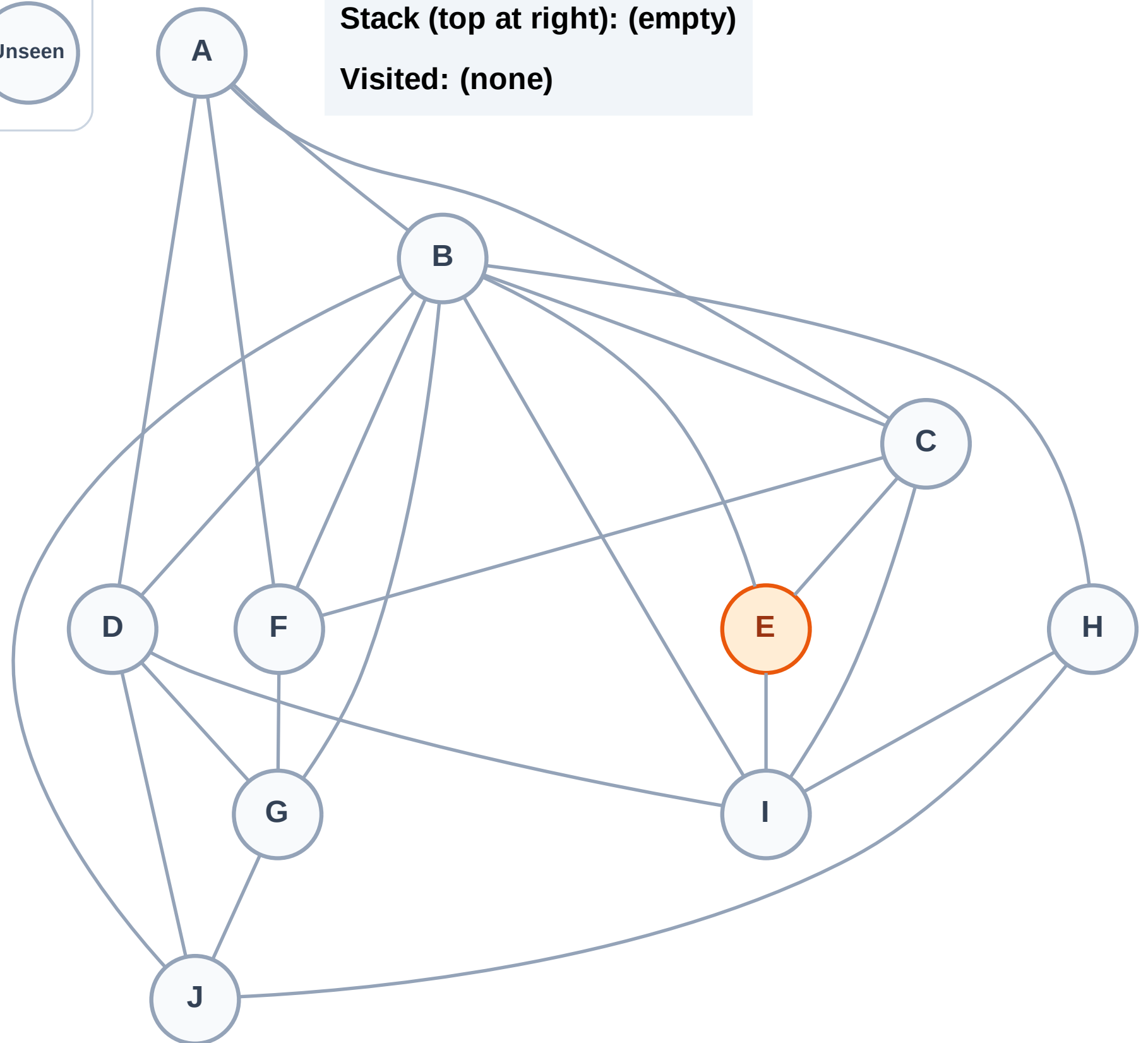
Step 2: Process node E

Legend



Stack (top at right): (empty)

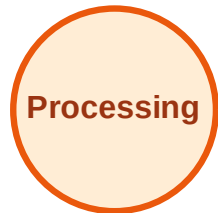
Visited: (none)



DFS Traversal

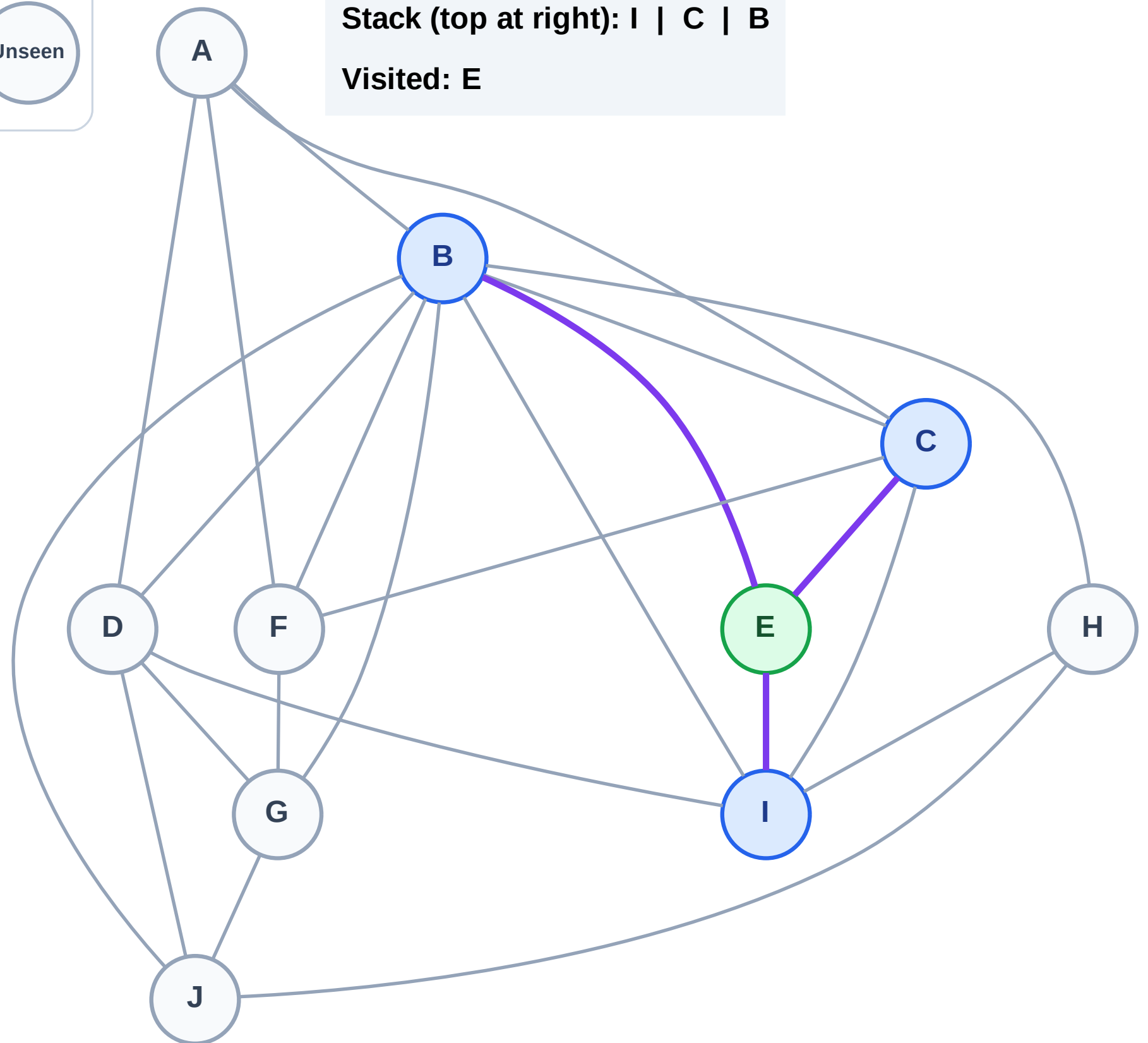
Step 3: Discover I, C, B from E

Legend



Stack (top at right): I | C | B

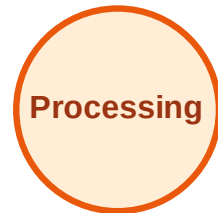
Visited: E



DFS Traversal

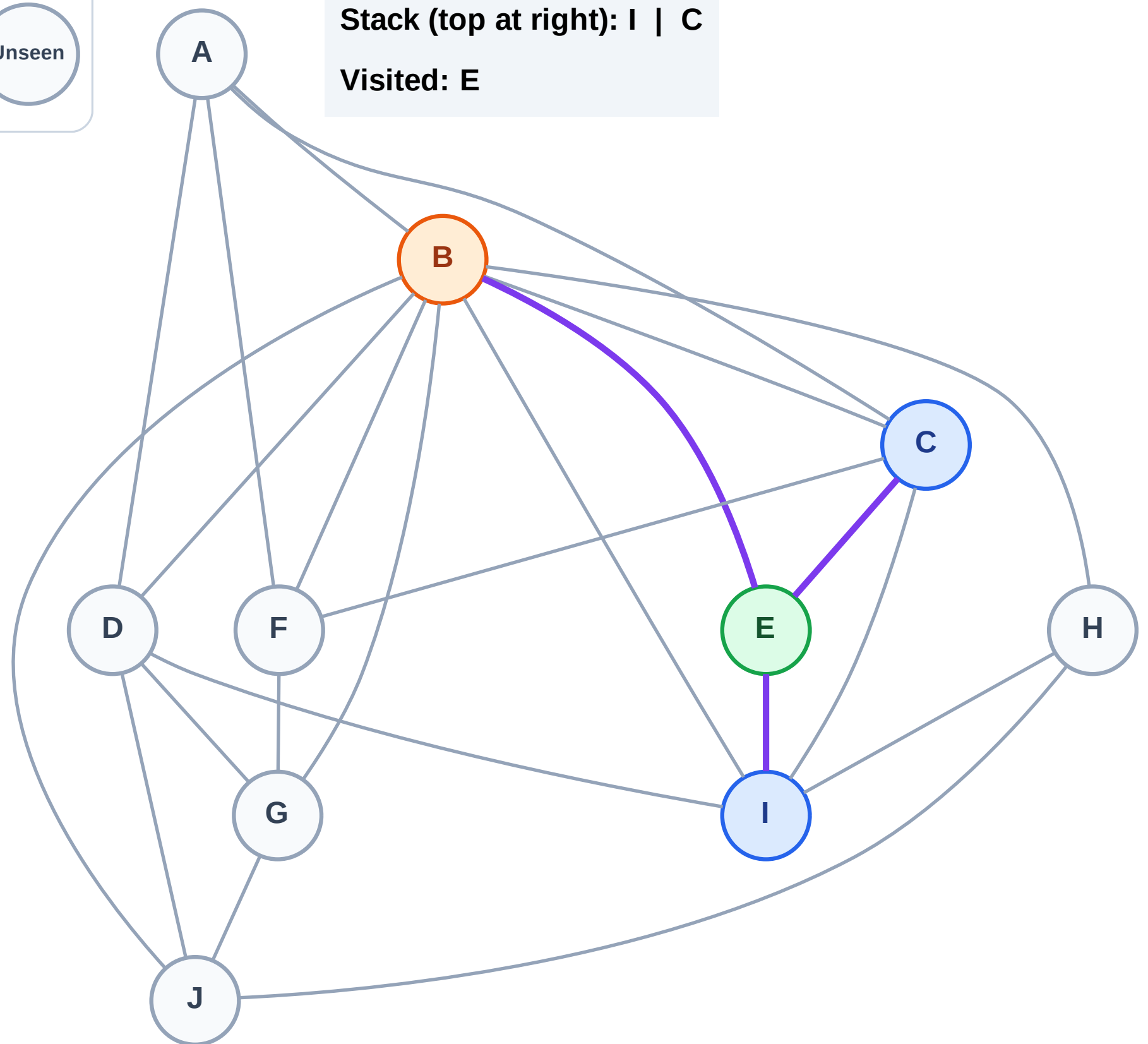
Step 4: Process node B

Legend



Stack (top at right): I | C

Visited: E



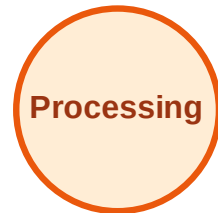
DFS Traversal

Step 5: Discover J, H, G, F, D, A from B

Legend



Complete



Processing



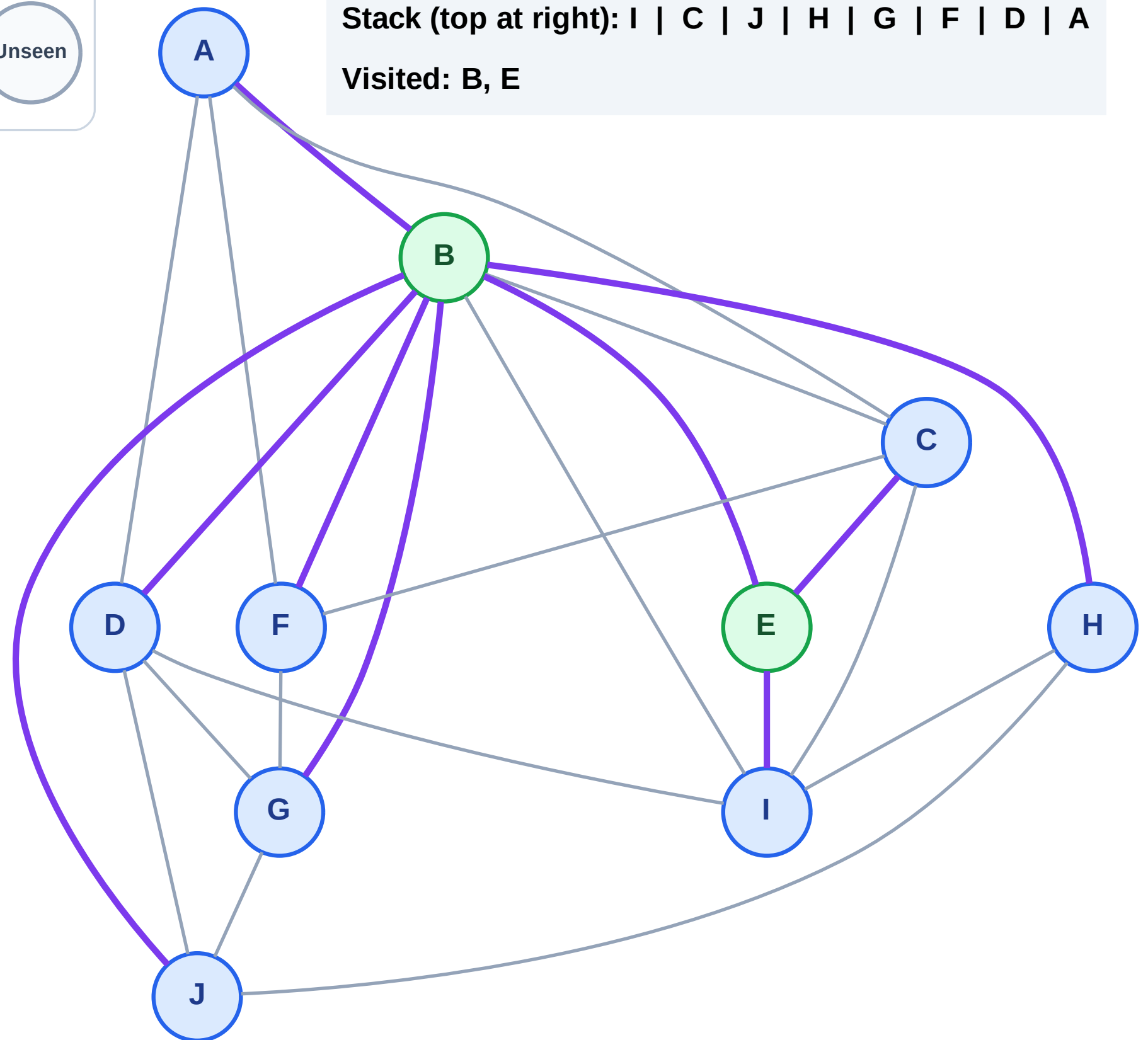
Discovered



Unseen

Stack (top at right): I | C | J | H | G | F | D | A

Visited: B, E



DFS Traversal

Step 6: Process node A

Legend

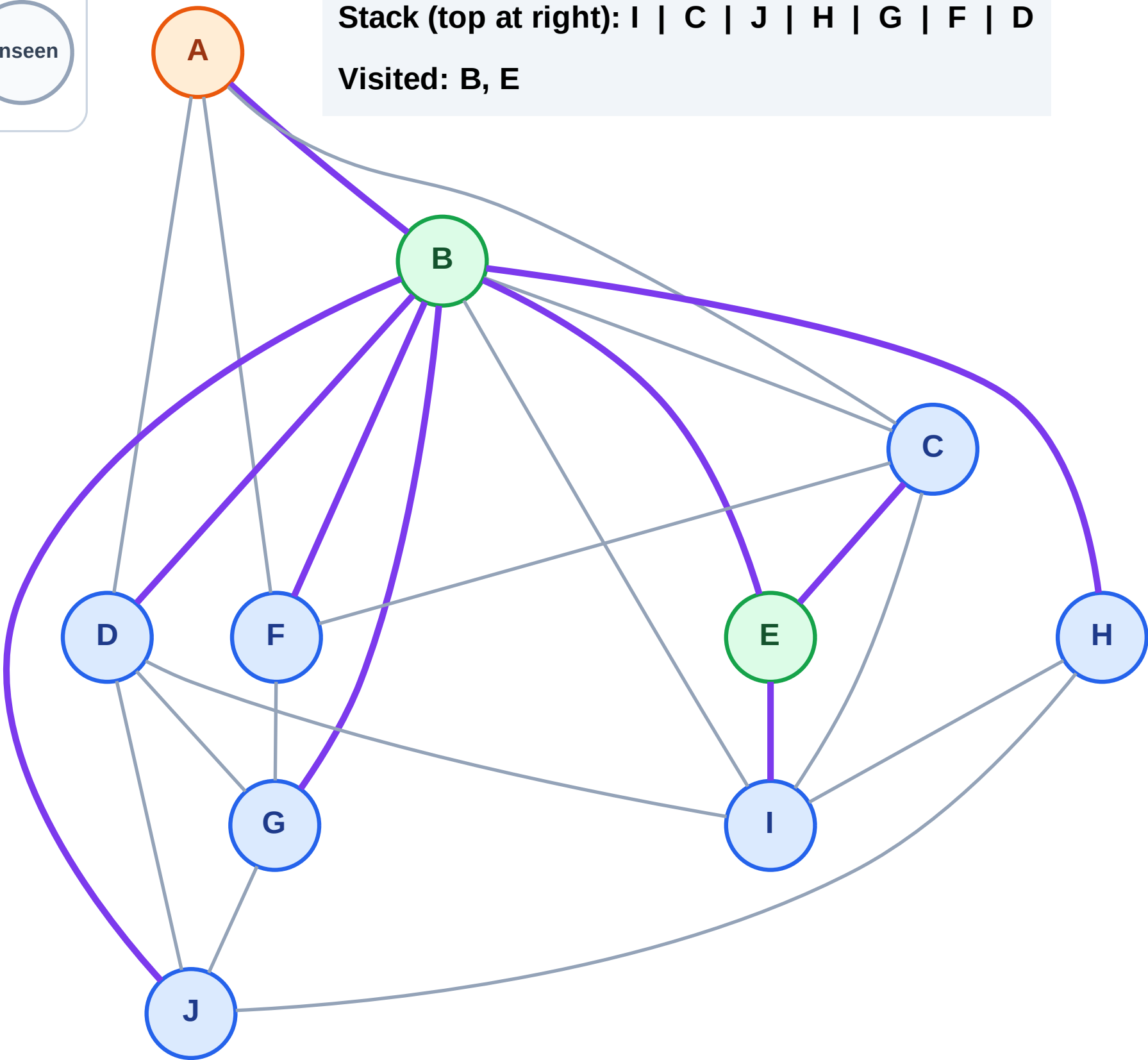
Complete

Processing

Discovered

Unseen

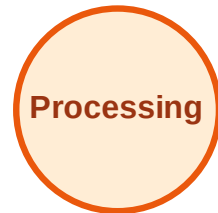
Stack (top at right): I | C | J | H | G | F | D
Visited: B, E



DFS Traversal

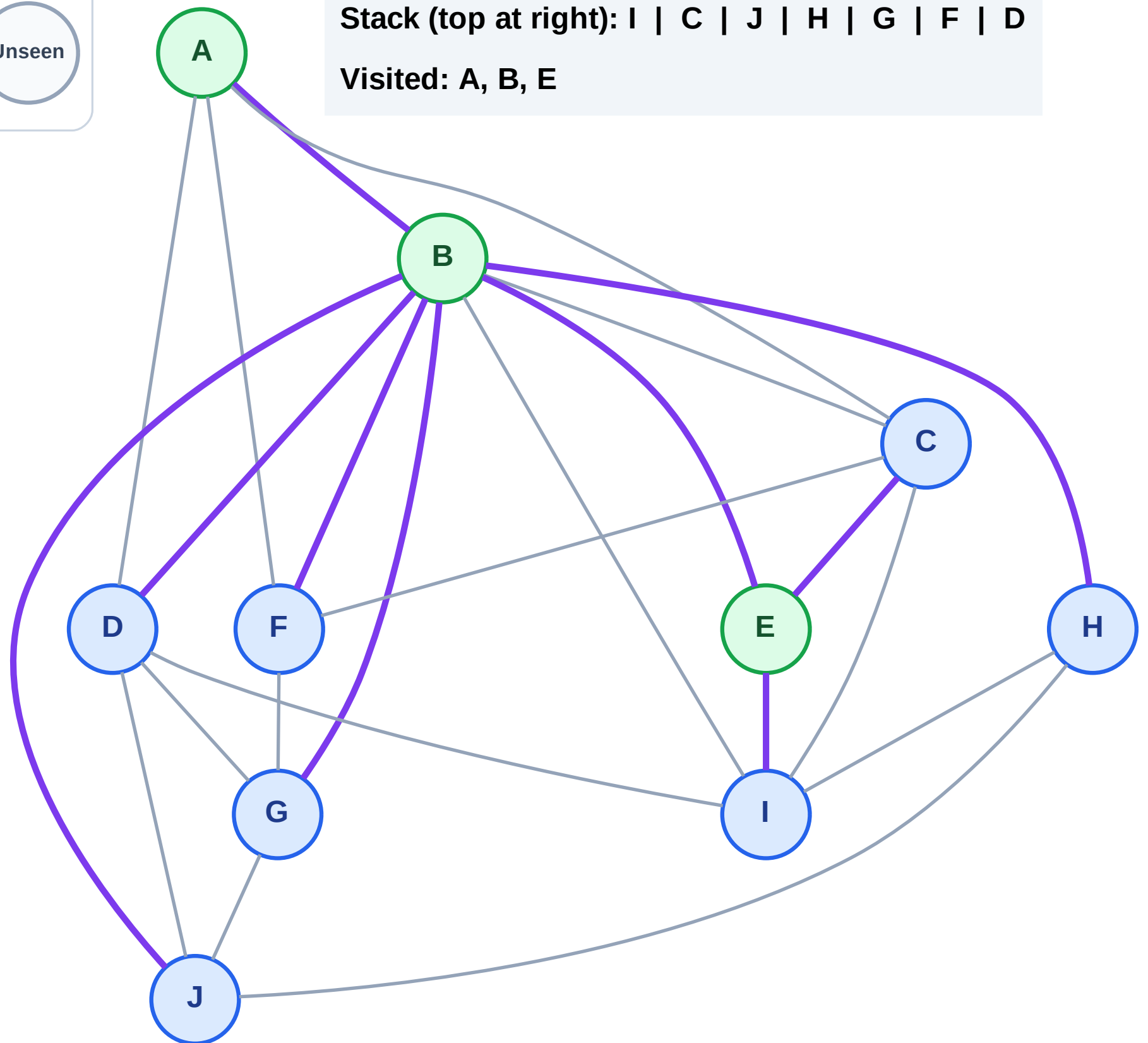
Step 7: No new neighbors from A

Legend



Stack (top at right): I | C | J | H | G | F | D

Visited: A, B, E



DFS Traversal

Step 8: Process node D

Legend

Complete

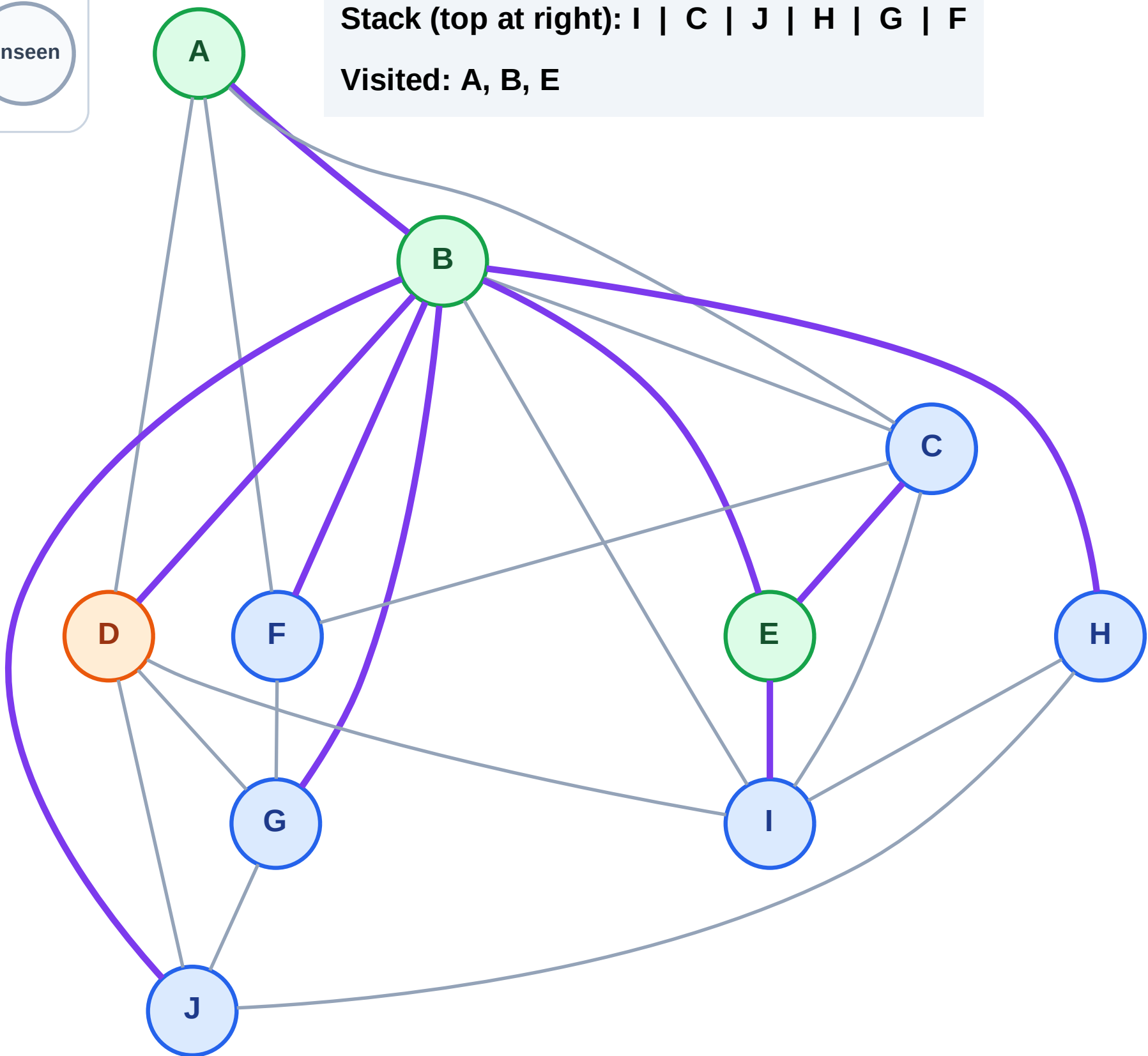
Processing

Discovered

Unseen

Stack (top at right): I | C | J | H | G | F

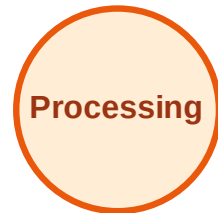
Visited: A, B, E



DFS Traversal

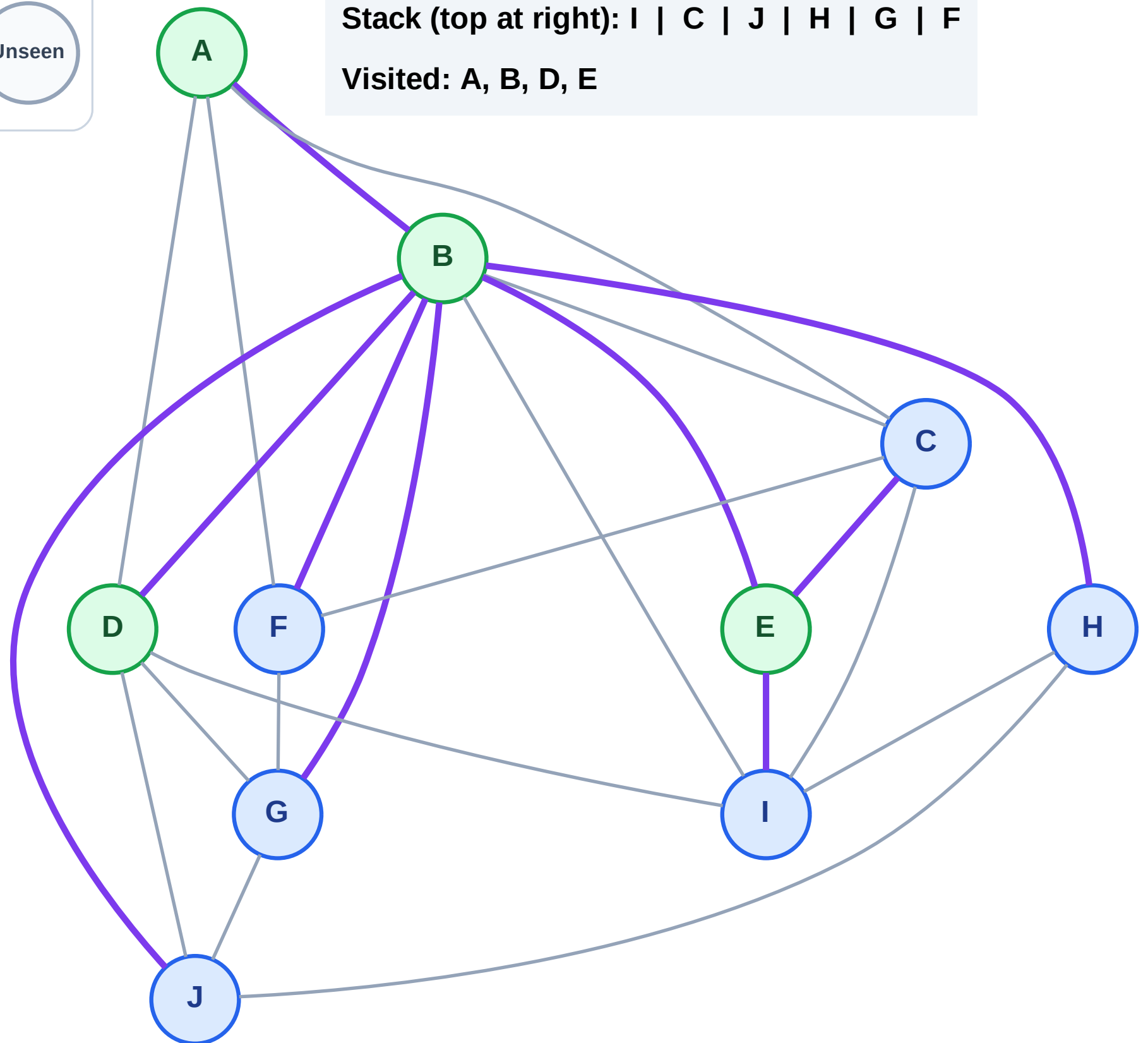
Step 9: No new neighbors from D

Legend



Stack (top at right): I | C | J | H | G | F

Visited: A, B, D, E



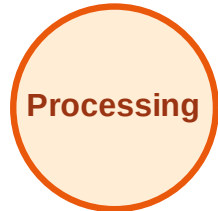
DFS Traversal

Step 10: Process node F

Legend



Complete



Processing



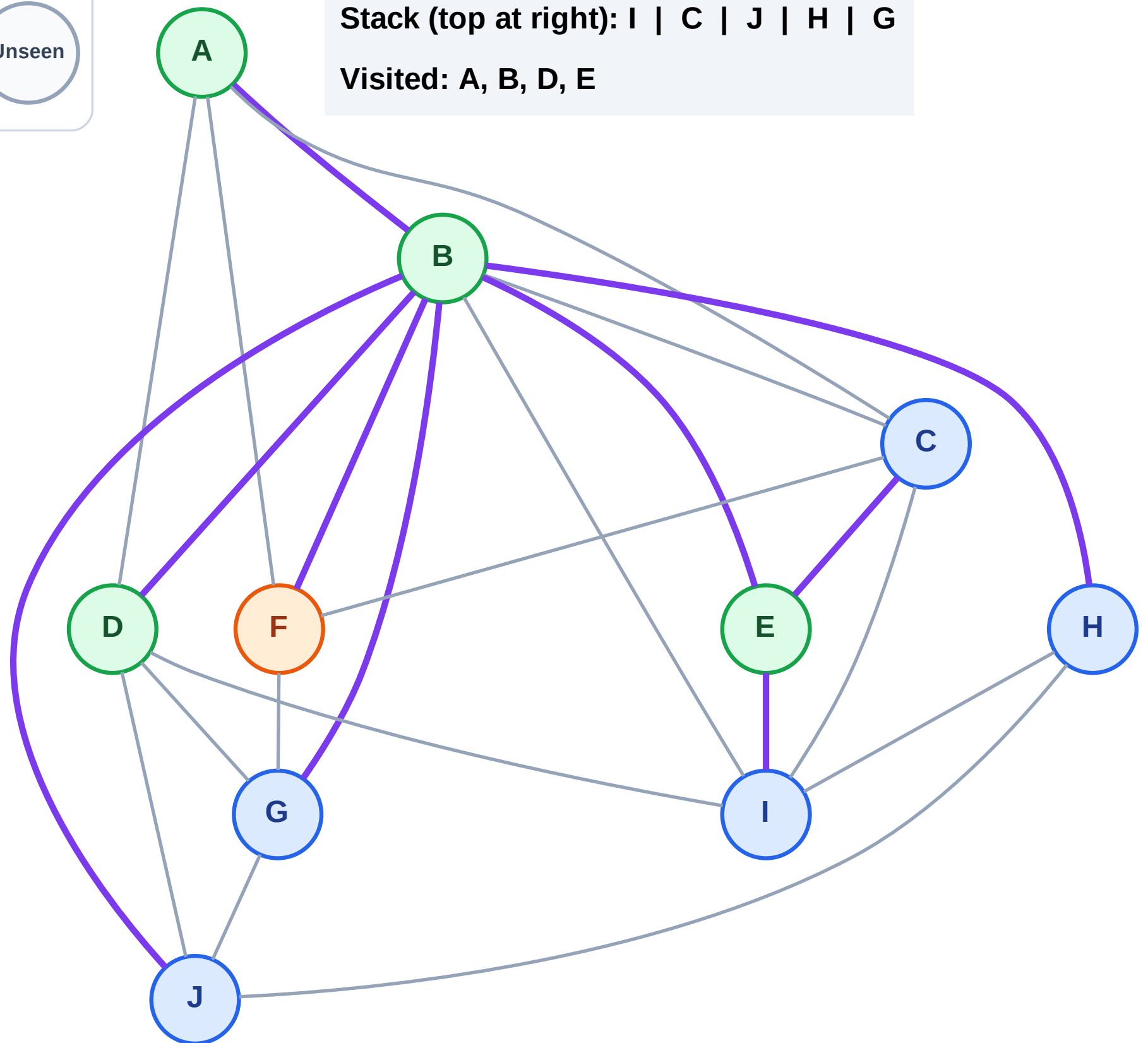
Discovered



Unseen

Stack (top at right): I | C | J | H | G

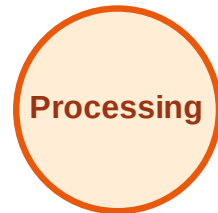
Visited: A, B, D, E



DFS Traversal

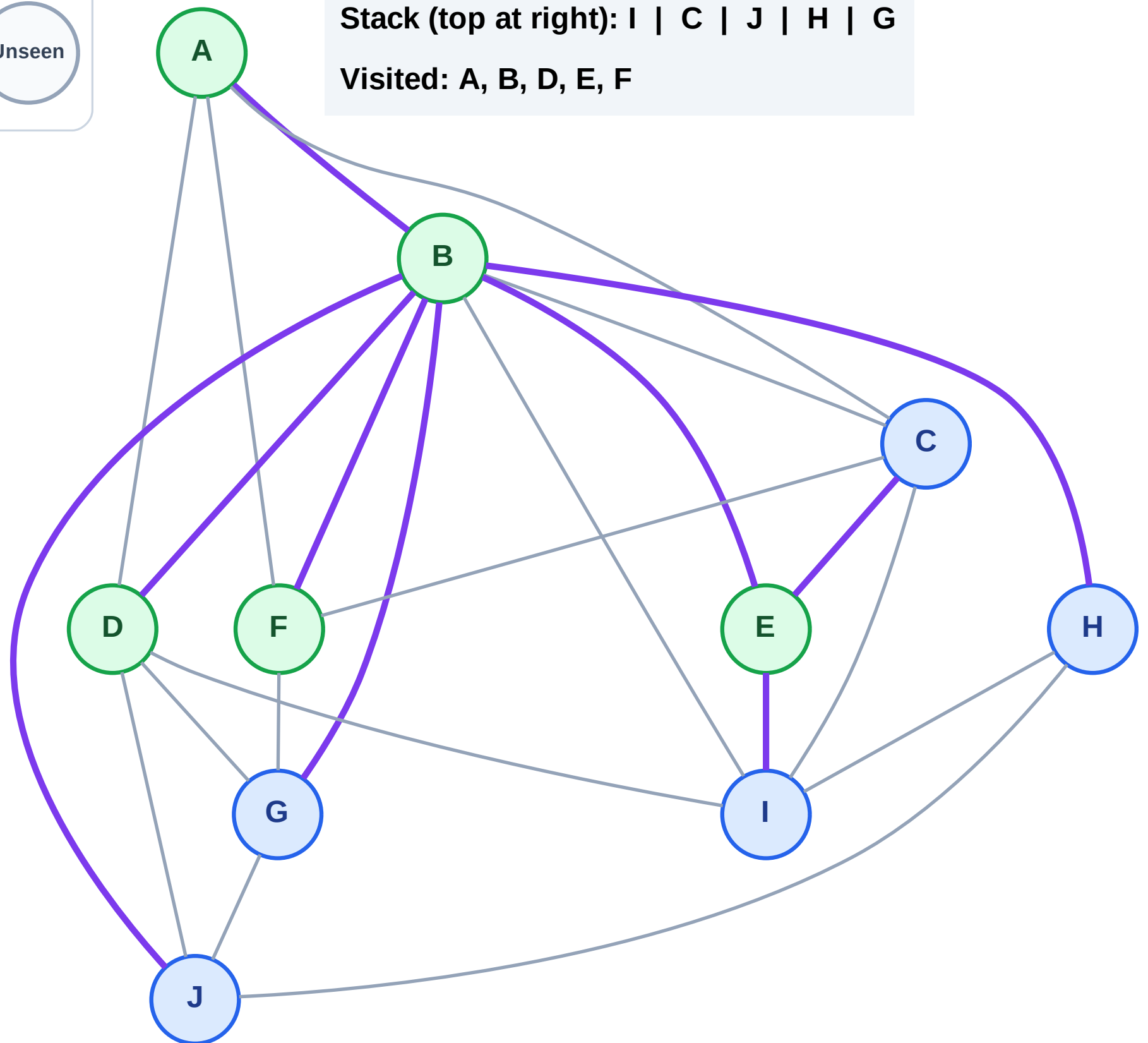
Step 11: No new neighbors from F

Legend



Stack (top at right): I | C | J | H | G

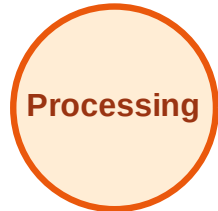
Visited: A, B, D, E, F



DFS Traversal

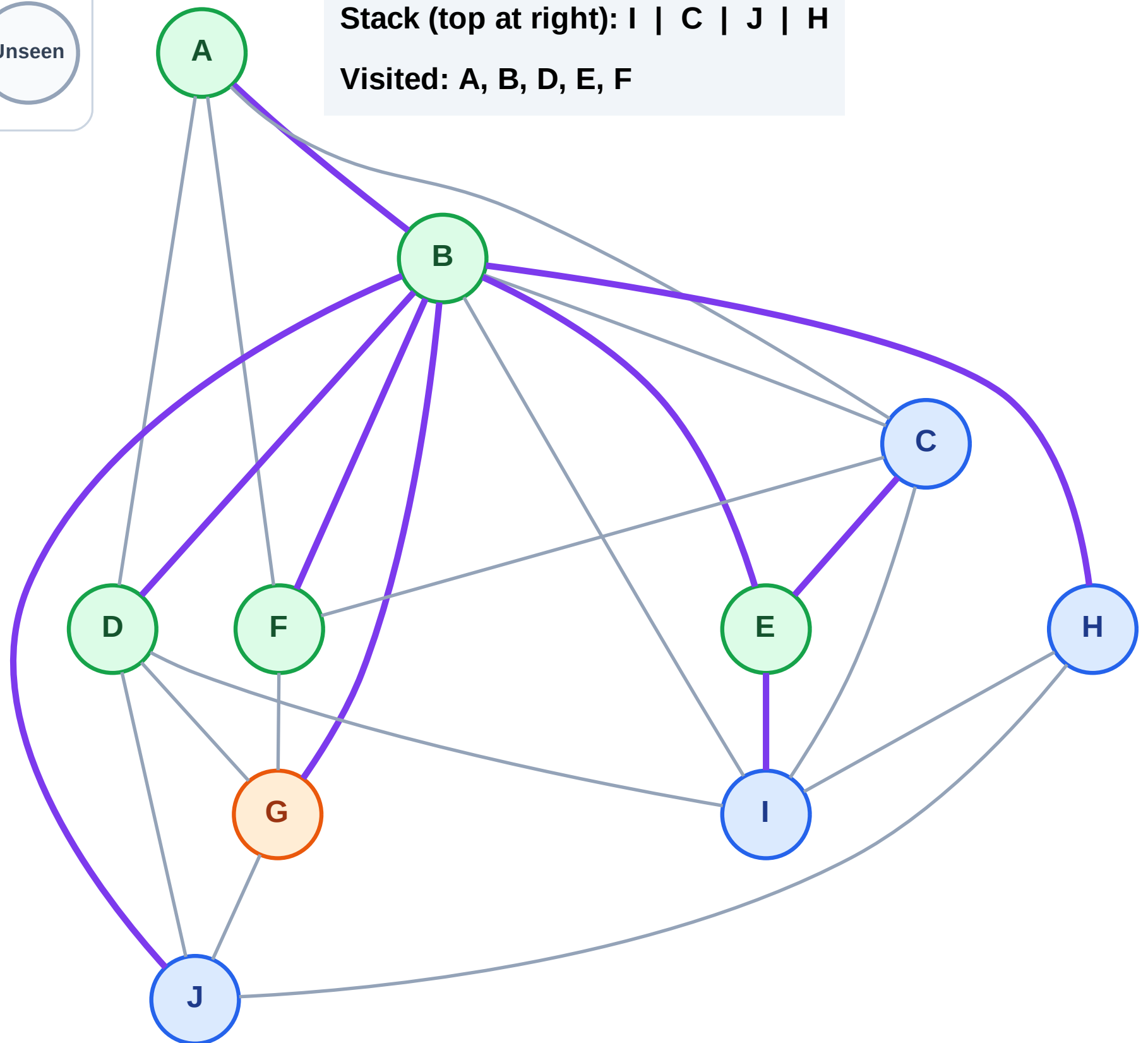
Step 12: Process node G

Legend



Stack (top at right): I | C | J | H

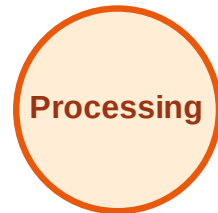
Visited: A, B, D, E, F



DFS Traversal

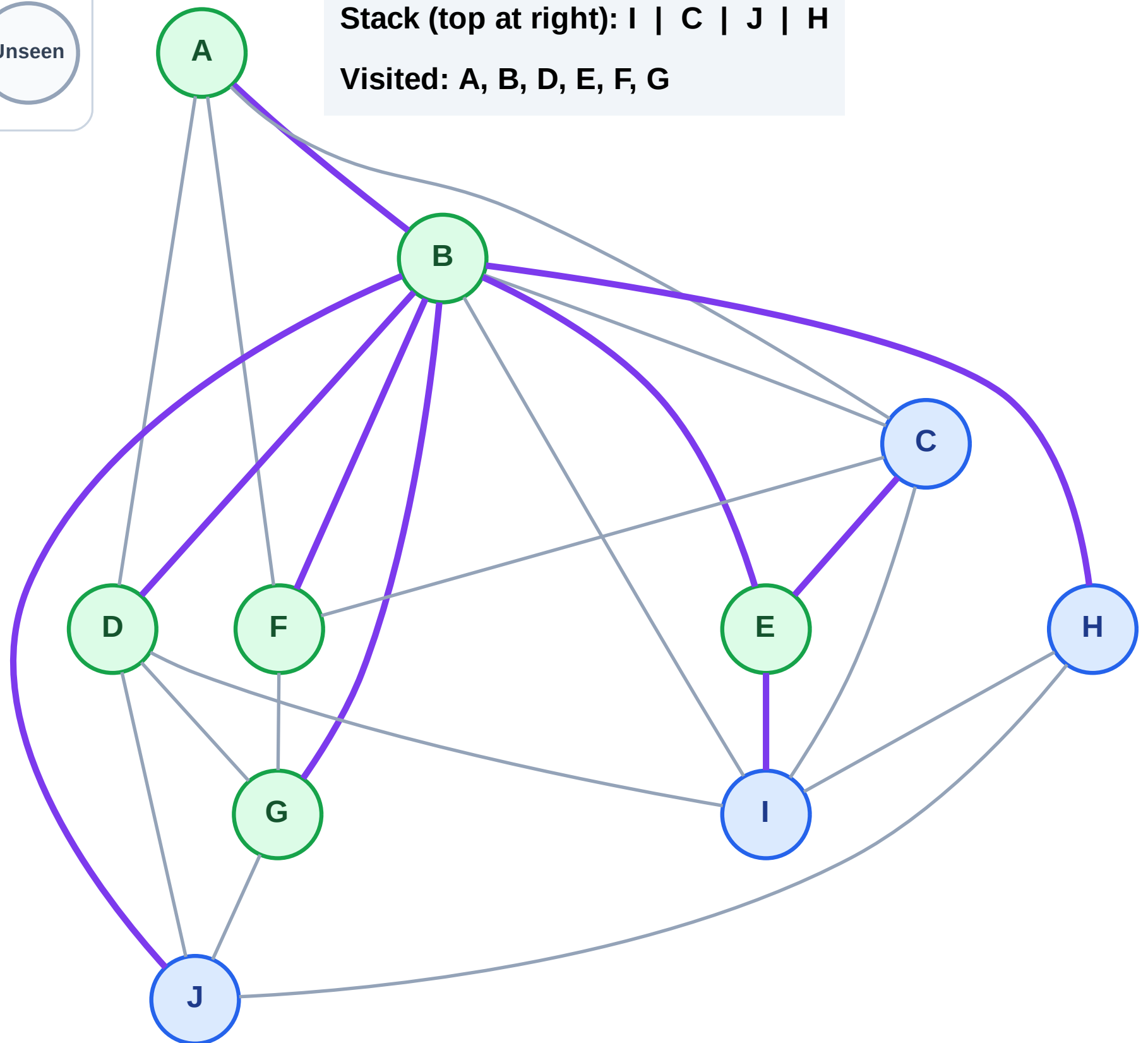
Step 13: No new neighbors from G

Legend



Stack (top at right): I | C | J | H

Visited: A, B, D, E, F, G



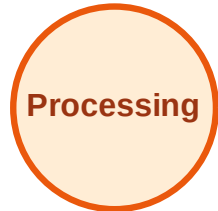
DFS Traversal

Step 14: Process node H

Legend



Complete



Processing



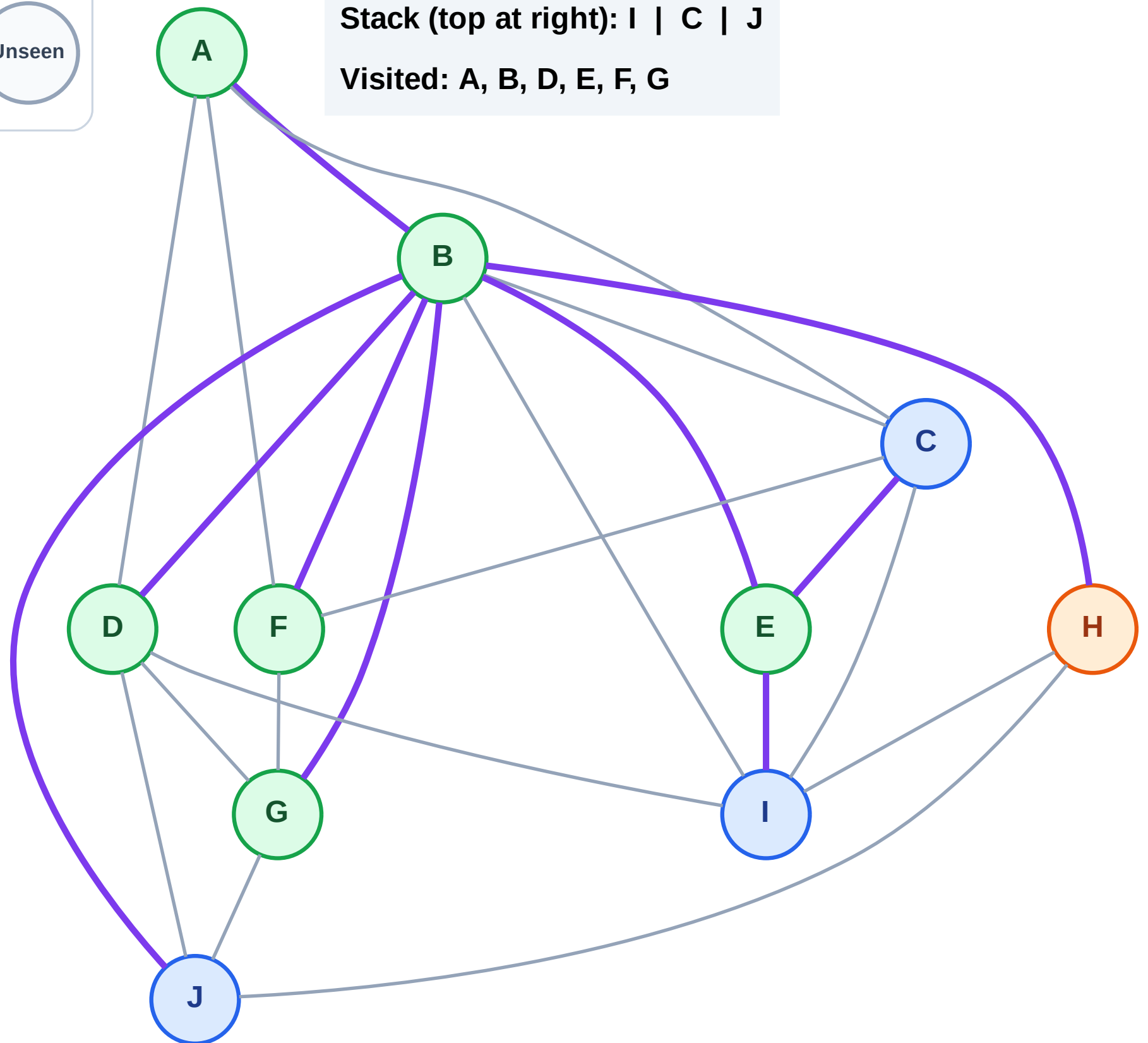
Discovered



Unseen

Stack (top at right): I | C | J

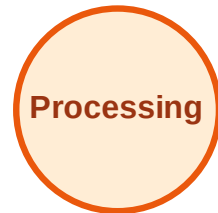
Visited: A, B, D, E, F, G



DFS Traversal

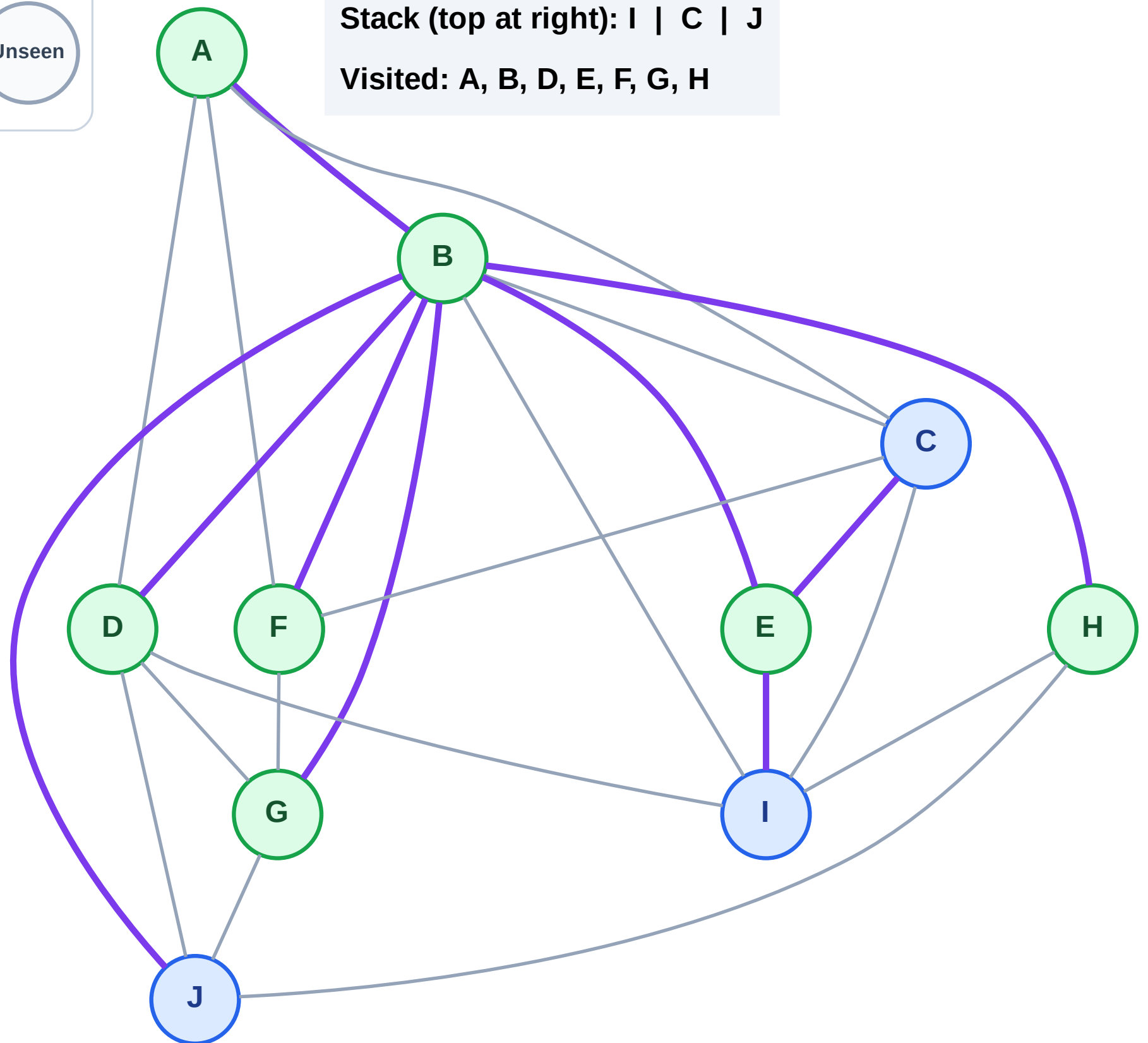
Step 15: No new neighbors from H

Legend



Stack (top at right): I | C | J

Visited: A, B, D, E, F, G, H



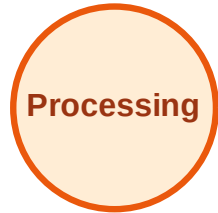
DFS Traversal

Step 16: Process node J

Legend



Complete



Processing



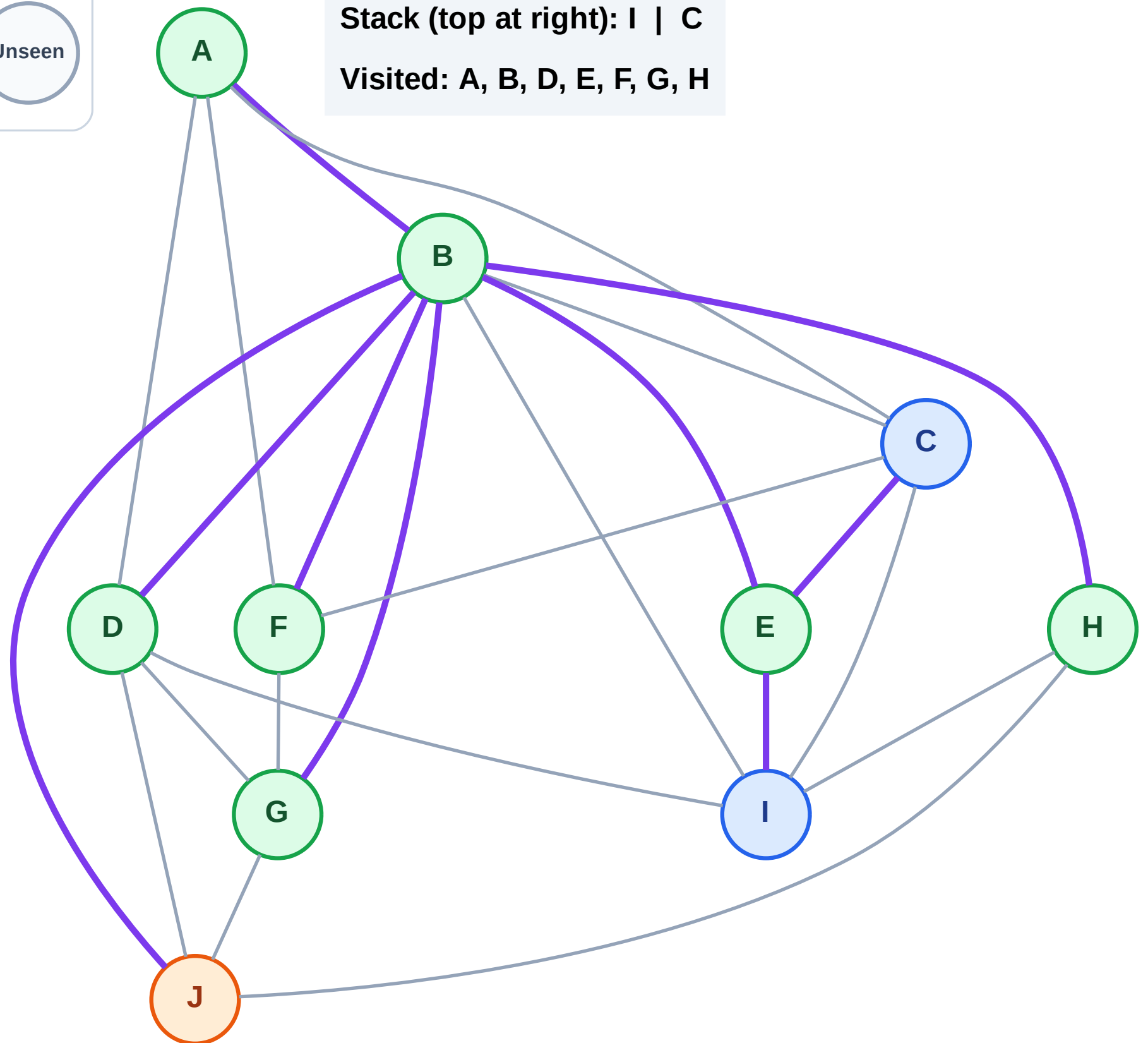
Discovered



Unseen

Stack (top at right): I | C

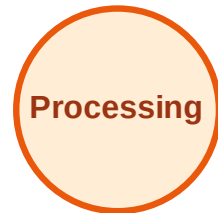
Visited: A, B, D, E, F, G, H



DFS Traversal

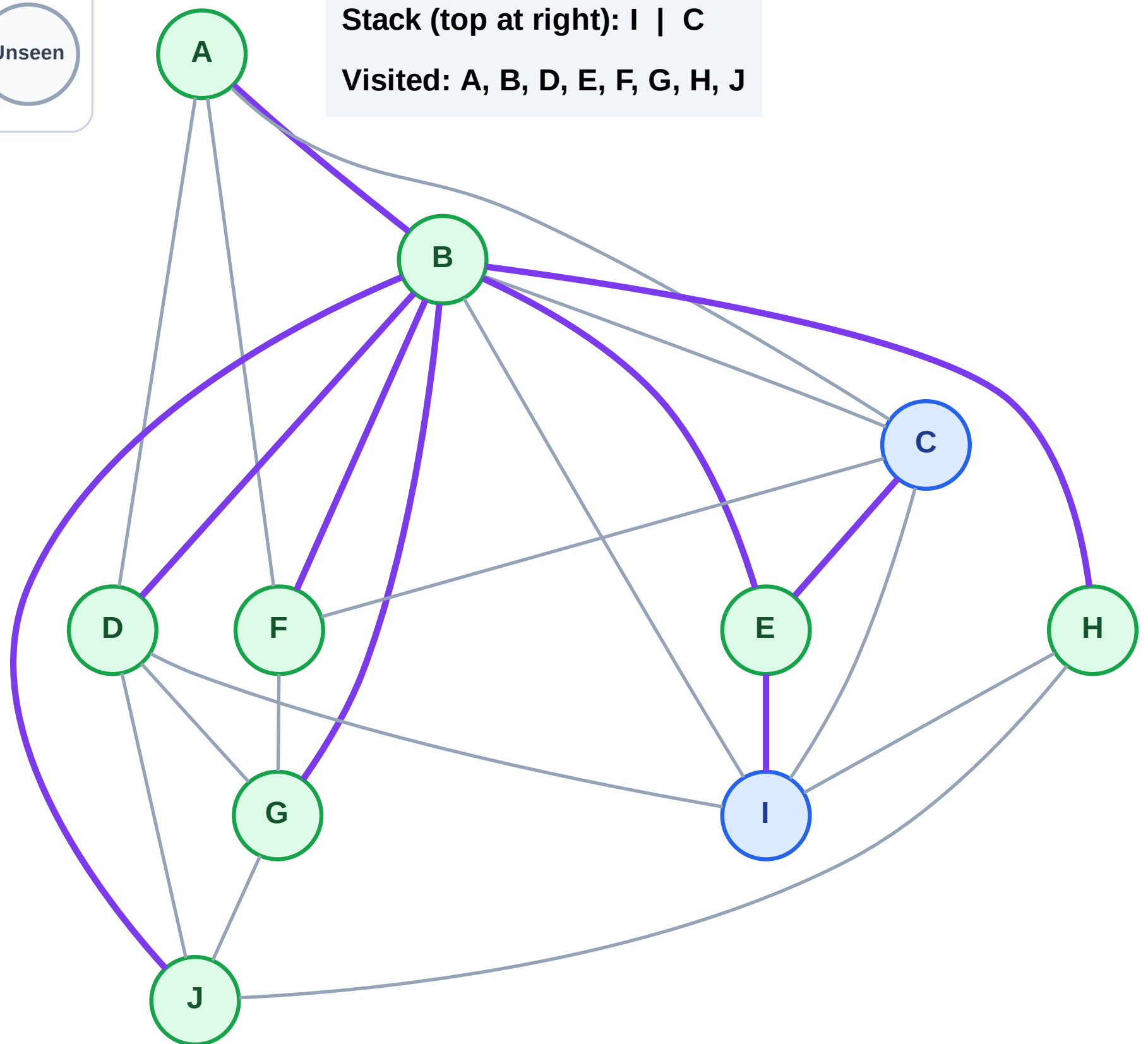
Step 17: No new neighbors from J

Legend



Stack (top at right): I | C

Visited: A, B, D, E, F, G, H, J



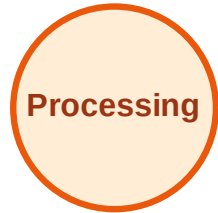
DFS Traversal

Step 18: Process node C

Legend



Complete



Processing



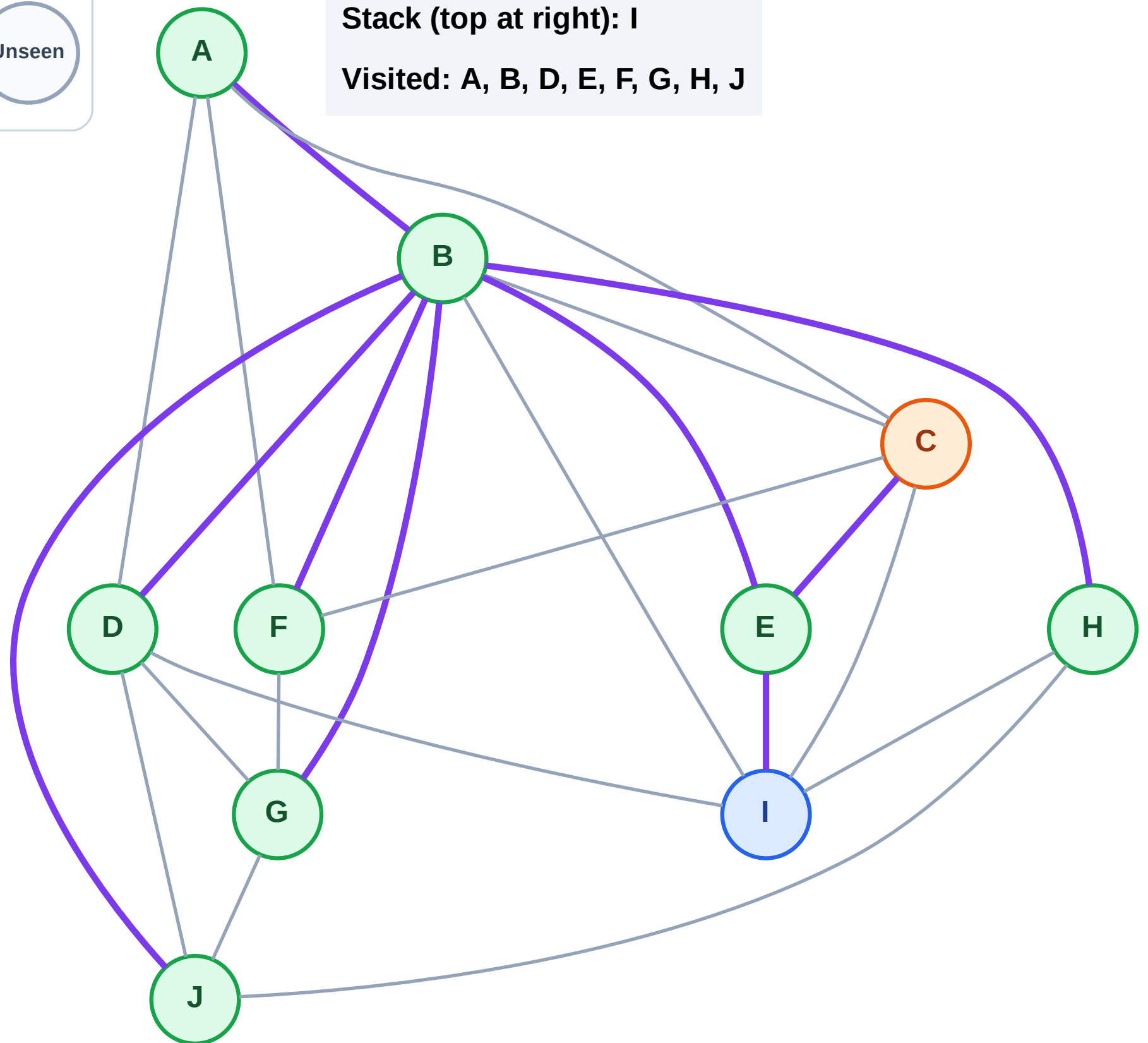
Discovered



Unseen

Stack (top at right): I

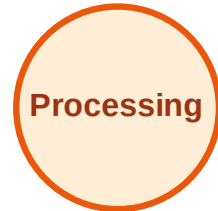
Visited: A, B, D, E, F, G, H, J



DFS Traversal

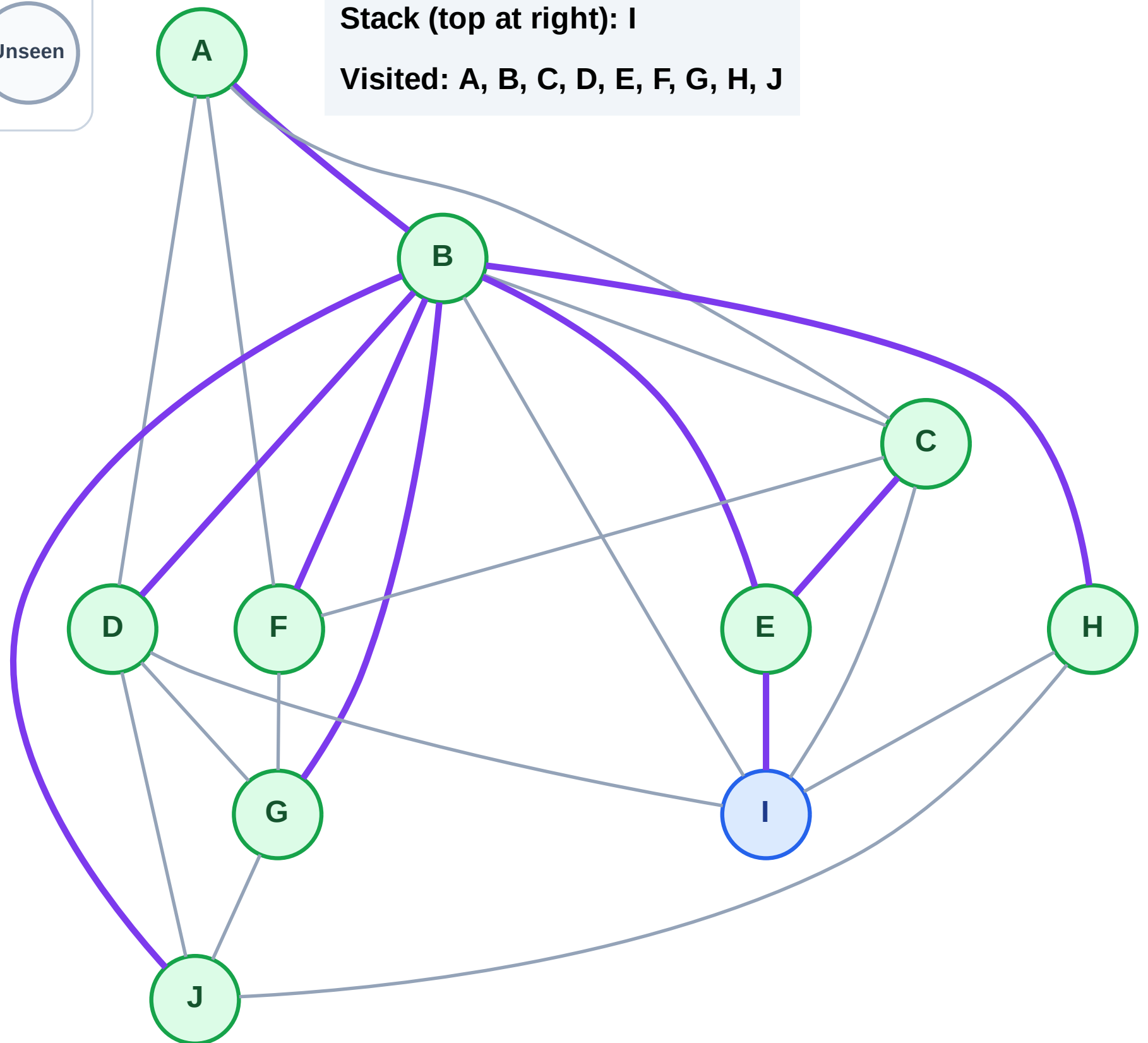
Step 19: No new neighbors from C

Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J



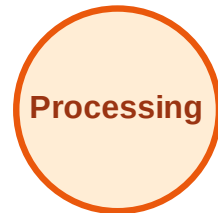
DFS Traversal

Step 20: Process node I

Legend



Complete



Processing



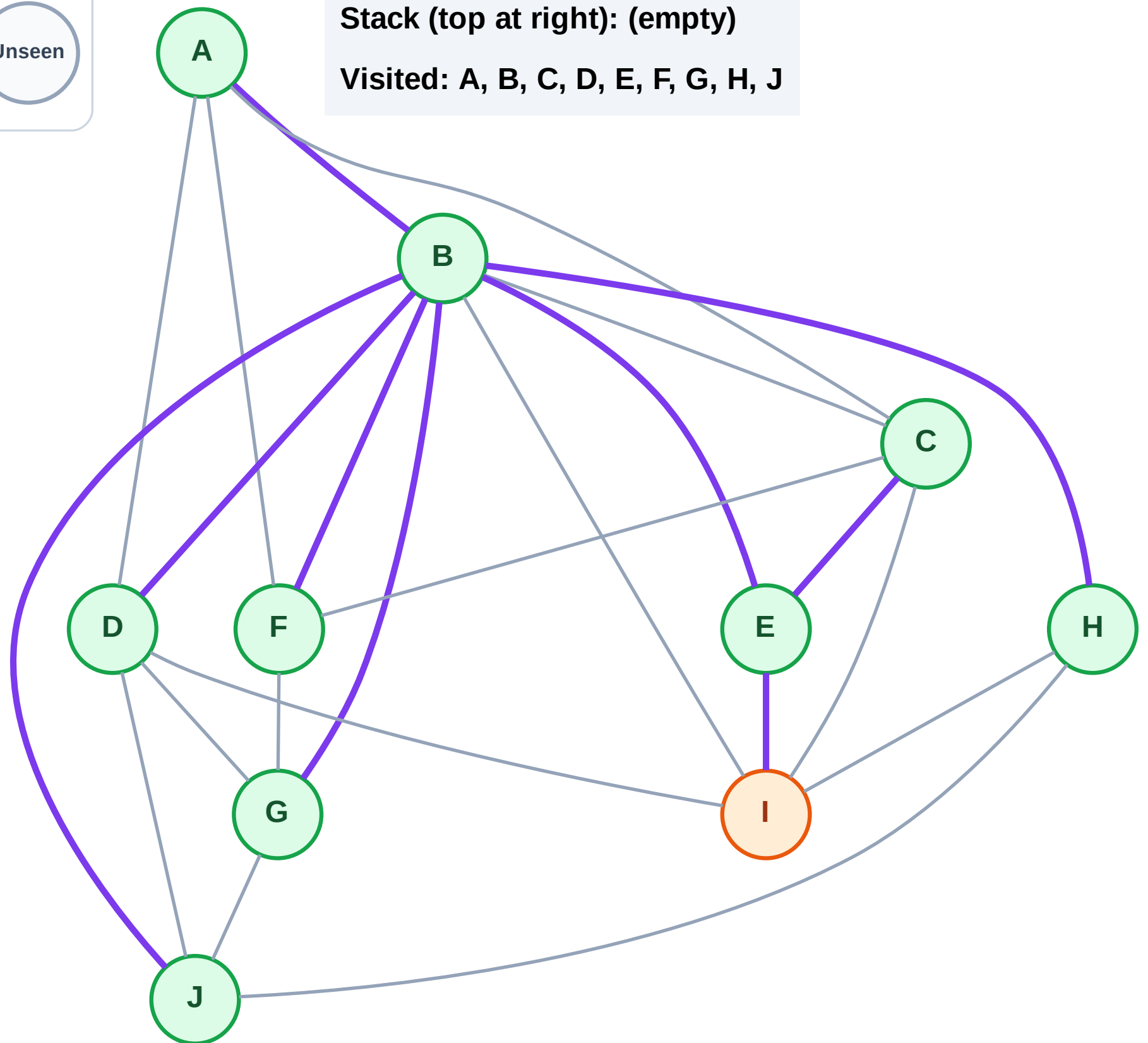
Discovered



Unseen

Stack (top at right): (empty)

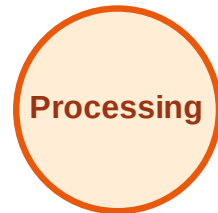
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

